scientific reports

OPEN

Utilizing deep learning models for early detection and classification of fruit diseases: towards sustainable agriculture and enhanced food quality

Ibrahim Alrashdi[1], Marwa Sharawi[2]✉, Ahmed M. Ali[3], Karam M. Sallam[4], Bilal Arain[5] & Mohamed Abdel-Basset[3]

Productivity and quality of food are crucial for populations around the world. However, food faces challenges due to the threats of fruit diseases, which lead to poor food quality. Therefore, early detection and classification of fruit diseases are important to help farmers detect and overcome these diseases, thereby improving food quality and productivity. One of the biggest challenges in the agriculture field is classifying and detecting fruit diseases using traditional manual visual grading. As a result, deep learning and computer vision models have emerged as new methods for visual grading, offering higher accuracy in classification and detection. This study proposes deep learning models for fruit disease detection and classification in the early stages. Five deep learning models are used: Convolutional Neural Network (CNN), DenseNet121, EfficientNetB3, Xception, and ResNet50. These models are applied to detect six types of fruit diseases, including orange, grape, mango, guava, apple, and banana plant diseases. Image preprocessing and data augmentation techniques were employed for image processing. The results show accuracies of 96.25%, 99.14%, 96.17%, 94.06%, 96.72%, and 99.33% for the CNN, EfficientNetB3, ResNet50, DenseNet121, ResNet50, and EfficientNetB3 models, respectively, for detecting orange, grape, mango, banana, guava, and apple plant diseases. We compared our models with other deep learning models, and the model that utilized image preprocessing and data augmentation techniques demonstrated higher accuracy and performance. We recommend the EfficientNetB3 model for fruit disease detection based on these results.

Keywords Fruits diseases detection, Deep learning, Agriculture, Sustainability, Fruit diseases, Food quality

The rapid growth of the global population necessitates a consistent supply of high-quality food to sustain this increasing demand. It is essential to enhance the productivity and quality of food production globally[1,2]. However, plant diseases pose significant challenges to achieving these goals, reducing yields and compromising food quality. Poor-quality food not only affects food security but also poses serious global health risks[3,4].

Detection and early classification of fruit diseases are essential to increase productivity, improve food quality, and improve population health. Fruit Disease Detection (FDD) is necessary for farmers to preserve the yield and enhance food quality[5]. Researchers use automated methods for FDD in the early phase, and the hardware systems are improved to easily capture higher quality and resolution images to detect disease in the early phase.

Image processing techniques play a crucial role in enhancing the quality of fruit images, making the detection of diseases more accurate and efficient. These methods are applied to preprocess images by adjusting key attributes such as size, color, and orientation[6,7]. By refining these characteristics, image processing ensures the data is optimized for subsequent analysis. One significant application is improving image resolution, vital for

[1]Department of Computer Science, College of Computer and Information Sciences, Jouf University, 2014 Sakaka, Saudi Arabia. [2]College of Engineering and Applied Sciences, American University of Kuwait, Safat, Kuwait. [3]Faculty of Computers and Informatics, Zagazig University, 44519 Zagazig, Sharqiyah, Oman. [4]Department of Computer Science, University of Sharjah, Sharjah, United Arab Emirates. [5]Department of Computer Engineering, University of Sharjah, Sharjah, United Arab Emirates. ✉email: mamostafa@auk.edu.kw

identifying even subtle signs of disease on fruits. High-resolution images enable the precise detection of small-scale features that may indicate the early stages of infection. Additionally, image processing helps reduce the size of excessively large images, ensuring they are manageable for computational models without losing essential details[8,9]. This preprocessing step not only streamlines the disease detection process but also enhances the accuracy and efficiency of the overall analysis, ultimately contributing to better decision-making in agricultural practices.

Deep learning (DL) has emerged as a powerful computational approach for handling unstructured data, including complex image datasets. Its ability to learn and adapt through convolutional techniques enables high accuracy and performance in image analysis tasks. One of the key strengths of DL is its capability to train neural networks using multilayered architectures, which allow the extraction of intricate features from data[10,11]. This feature extraction process is essential for identifying image patterns, making DL an ideal choice for FDD. DL model was applied in different application to obtain higher accuracy and performance[12,13].

This paper applies five state-of-the-art deep learning models: Convolutional Neural Network (CNN), DenseNet121, EfficientNet B3, Xception, and ResNet50. These models are well known to enhance the accuracy and efficiency of FDD. Each model brings unique strengths and architectural innovations that contribute to their effectiveness in identifying and classifying diseases in fruit images. Their combined application ensures robust performance across diverse datasets and disease types. CNN model was applied for different classification tasks[14–18].

The Convolutional Neural Network (CNN), a foundational model in this paper, excels in image recognition and processing tasks[19]. Its layered architecture, comprising convolutional, pooling, and fully connected layers, enables the extraction and classification of patterns within images. CNN is widely recognized for its ability to identify complex visual features, making it an essential tool for FDD, particularly in detecting subtle disease symptoms[20,21].

DenseNet121, a densely connected convolutional network, enhances feature reuse by directly connecting each layer to every other subsequent layer[22]. This dense connectivity reduces the number of parameters, mitigates the risk of overfitting, and improves model efficiency. DenseNet121 is particularly effective in scenarios requiring detailed feature extraction, as it preserves spatial information critical for accurate disease detection in fruits[23].

EfficientNet B3 adopts a compound scaling approach, simultaneously optimizing model depth, width, and resolution[24]. This design achieves superior accuracy with fewer parameters and lower computational costs than traditional models. EfficientNet B3 is well-suited for FDD due to its ability to balance precision and efficiency, making it ideal for applications with limited computational resources[25].

Xception[26] improves upon the Inception model[27] using depthwise separable convolutions, drastically reducing computational complexity while maintaining high accuracy. Its architecture is adept at handling large and complex datasets, effectively identifying various disease patterns in high-resolution fruit images[28]. Xception's flexibility and scalability contribute to its strong performance in this domain.

ResNet50[29], a deep residual network, addresses the challenge of vanishing gradients in deep networks by introducing shortcut connections. These connections enable the model to learn residual mappings instead of direct mappings, significantly improving training efficiency and accuracy. ResNet50 is particularly effective in extracting hierarchical features from images, allowing for precisely identifying disease symptoms across different fruit types[30].

This paper presents a comprehensive approach to fruit disease detection by applying the five different DL models. While CNN provides a solid foundation for image analysis, models like DenseNet121 and EfficientNet B3 optimize efficiency and accuracy. Meanwhile, Xception and ResNet50 offer advanced feature extraction capabilities and generalization ability across diverse datasets. By leveraging the unique strengths of each model, this paper achieves a robust and reliable system for detecting and classifying fruit diseases, ultimately contributing to improved agricultural productivity and food quality.

To this end, the main contributions of this paper are as follows:

- This paper applies five advanced DL models for effectively detecting and classifying fruit diseases. The models were trained and evaluated on six case studies involving different fruits: apple, guava, orange, banana, grape, and mango.
- Image preprocessing and data augmentation techniques were utilized to prepare and enhance the input image data, improving the overall accuracy and reliability of the DL models.
- The evaluations demonstrated superior accuracy and performance compared to existing results in the literature for classifying and detecting fruit diseases. A comprehensive comparison is provided for the readers using different datasets.
- The results demonstrated in this paper provide practical applications for farmers, enabling the early detection of fruit diseases and enhancing agricultural product quality.

The organization of this study is as follows: "Literature review" provides a review of the literature on fruit disease classification and detection. "Methodology" outlines the methodology used in this study. "Results and discussion" presents the six datasets, the results of the deep learning models, a discussion of the results, and a comparative analysis with other DL models. "Challenges in fruit plant disease management and theirimplications for agriculture" highlights the challenges faced by farmers in managing fruit diseases. "Conclusions and future directions" concludes the study.

Literature review

This section provides a literature review on fruit disease detection using machine learning models.

Dhiman et al.[31] proposed a DL algorithm for fruit disease prediction. They utilized a deep neural network (DNN) for detection and classification, along with transfer learning using VGGNet for multi-class classification tasks. Their model achieved 99% accuracy in low and high-severity cases, 98% and 96% in healthy datasets, and 97% in medium-severity cases. This demonstrates the effectiveness of their model in fruit disease detection and classification. Nasir et al.[32] proposed a DL framework for detecting and classifying fruit diseases. They employed a CNN model to extract features from the dataset and fine-tuned the model using VGG19 for fruit disease classification. Their model achieved an accuracy of 99.6%.

Ashok and Vinod[33] introduced a pre-trained deep CNN algorithm to identify mango fruit diseases. Using transfer learning, they trained their model on a mango disease dataset, which was integrated into a smartphone application. The model achieved 98.6% accuracy on training data and 96.4% on the validation dataset. Gill et al.[34] proposed a framework for classifying, segmenting, and recognizing fruit diseases using DL models integrated with a type-2 fuzzy model, CNN, recurrent neural network (RNN), and long short-term memory (LSTM) models. Their model demonstrated high accuracy and provided quantitative analysis results. Assunção et al.[35] developed DL models for classifying and detecting fruit diseases using CNN for feature extraction. Their model, which can be run on mobile devices, successfully classified three types of peach diseases and healthy fruits, achieving a 96% F1 score and no misclassifications.

Khattak et al.[36] proposed DL models for fruit disease classification and detection. They used CNN for classifying healthy fruits and diseased fruits, with feature extraction performed through various CNN layers. Their model achieved 94.55% accuracy and outperformed other models across multiple evaluation metrics. Gómez-Flores et al.[37] employed DL approaches for fruit disease detection. They used a CNN model for feature extraction from images, applying it to orange fruit diseases. Their model was trained using 953 images. Banerjee et al.[38] combined CNN and random forest models for orange disease prediction, achieving accuracy between 93.11% and 94.67%. Their model demonstrated superior performance across various disease classes. Dhiman et al.[39] presented a DL model for fruit disease prediction that combined CNN with LSTM, utilizing feature extraction and fusion techniques for image recognition. Their model achieved 97.18% accuracy on plant disease datasets.

Mojumdar and Chakraborty[40] introduced an automated fruit disease detection model applied to an orange disease dataset. They used image processing techniques to assess the severity of the disease, the K-means algorithm for image segmentation, and support vector machine (SVM) for classification. Their model achieved 82.3% accuracy. Saha et al.[41] developed a methodology for fruit disease classification and detection, using DL models to recognize images of orange fruit diseases. Their model achieved 93.21% accuracy and aims to automate the classification and detection process, ultimately improving economic outcomes and farmers' quality of life. Malathy et al.[42] used a DL model for fruit disease detection and classification, employing a CNN model for feature extraction. Their model achieved 97% accuracy and can assist farmers in improving crop growth in the future. Kukreja and Dhiman[43] introduced a CNN model for fruit disease detection, incorporating image processing and data augmentation techniques. Their model achieved 89.1% accuracy, highlighting the significance of preprocessing methods and data augmentation in improving fruit disease detection performance.

Awate et al.[44] proposed a K-means algorithm for image segmentation in fruit disease detection, coupled with an artificial neural network (ANN) model for classification. Their model supports the automation of farming processes. Gaikwad et al.[45] designed a system for fruit disease detection and classification using SVM for classification and the K-means algorithm for image segmentation. Their model provided accurate detection and automated classification of fruit diseases. Agarwal et al.[46] developed an automated system for fruit disease detection and classification, using K-means for image segmentation and SVM for classification. Their model achieved an accuracy of 98.39%. The literature review highlights various approaches and techniques in the field of fruit disease detection using machine learning, demonstrating the effectiveness and potential of deep learning models in addressing agricultural challenges.

Economic applications of deep learning in agriculture

- Early detection of fruit diseases helps prevent widespread crop damage, reducing overall losses.
- Minimizing spoilage and waste from undetected diseases leads to higher yields and better utilization of available resources.
- Deep learning models automate disease detection, reducing the need for labor-intensive manual inspections.
- Efficient use of pesticides and fertilizers by targeting treatments only to infected crops, leading to cost reduction.
- Early intervention through accurate disease classification allows farmers to take timely corrective actions, which can boost crop yields and overall productivity.
- By minimizing crop losses, farmers can maintain consistent and higher-quality produce, improving their income.
- Better quality control through early disease detection ensures that crops meet market and export standards, expanding market access and improving competitiveness in global agricultural markets.
- Reducing dependence on harmful chemicals (pesticides, fungicides) helps promote environmentally sustainable farming practices, which in turn can lead to long-term economic benefits for both farmers and the environment.
- Sustainable practices help maintain soil health and biodiversity, reducing the cost of soil restoration and other environmental damage in the long run.
- Deep learning tools can be adapted to farms of all sizes, offering affordable disease detection solutions to smallholder farmers, thereby boosting their profitability and productivity.

- Large-scale farms can integrate these models into their existing agricultural management systems, improving operational efficiency and reducing costs.
- Early detection leads to more reliable harvest forecasts, improving the efficiency of the entire supply chain (from farmers to distributors to consumers).

Methodology

This section outlines the methodology used in this study, as illustrated in Fig. 1. The process involves several key steps, including image processing, data processing, and training, with a total of five models applied across all the case studies.

- Image data augmentation (IDA)

IDA is used to solve the problem of limited ML training data. It is used to modify the images of the training dataset. It is used to increase the number of images in the training dataset. It is used to reduce the overfitting problem in ML problems. There are two types of IDA: one is used to change the location of pixels, and the second is used to change the values of image pixels. There are methods for changing the locations of pixels like rotation, flipping, translation, cropping, and scaling[47].

- Image pre-processing (IPP)

IPP is used in deep learning models to process the images. IPP is used to format the images before using them by the trained model. There are various methods in IPP, such as resizing, color corrections, and orienting. IPP minimizes the training time and speeds up the prediction results. IPP is used to reduce the size of images to reduce the training time without removing important features and to obtain higher accuracy[48].

CNN

CNN model is a class of neural networks. It is used to process images to make predictions and produce output. CNN has three main elements ConV layer, pooling layer, and fully connected layer.

DenseNet121 (DN)

DenseNet is an extension of the CNN model used for image classification problems because it obtains higher accuracy and performance[49]. It was introduced in 2016 as a CNN model. DenseNet structure achieved better performance in the image categorization process. It connects each layer with each layer in feedforward. It has two main functions such as bottleneck layers and dense block architecture. Bottleneck layers are used to minimize the number of parameters without minimizing the number of features. The dense block is used for connecting each layer in feedforward networks. The architecture of DenseNet121 is shown in Fig. 2. DenseNet architecture are follows:

- Bottleneck layer (BL)

BL has two convolutional layers (ConV) and a batch normalization layer in between them. The first ConV is used to minimize the number of features maps and the second ConV is used to produce the output feature maps.

- Transition Layer (TL)

TL has three layers such as batch normalization, ConV, and average pooling layer. Minimization of the number of feature maps by batch normalization, ConV, and average pooling layer used to minimize the spatial dimensionality of features map.

EfficientNet (EN)

EN used a scaling method based on a cutting-edge CNN algorithm[50]. It is used to obtain a better outcome by simple compound coefficients. It scales every dimension, such as resolution, depth, and breath; this is opposed to convolutional algorithms. We can improve the efficiency and performance of the EN model in the dataset by balancing every available source and enhancing each size scaled[51]. Figure 3 shows the structure of EN. The depth, width, and resolution of EN can be computed as:

$$depth : d = a \propto \tag{1}$$

$$width : w = b \propto \tag{2}$$

$$resolution : r = c \propto \tag{3}$$

where $\propto$ used to scale the model or regulate the access to the sources. a, b, c are used to search and define how extra sources are gathered to depth, width, and resolution.

Xception (XC)

XC model is an adaption of the Inception model and was proposed in 2016. It is referring to extreme inception. It is an extension of the CNN model used for image classification[52]. Figure 4 shows the structure of XC. There are four components of XC:

- Depth-wise separable convolutions

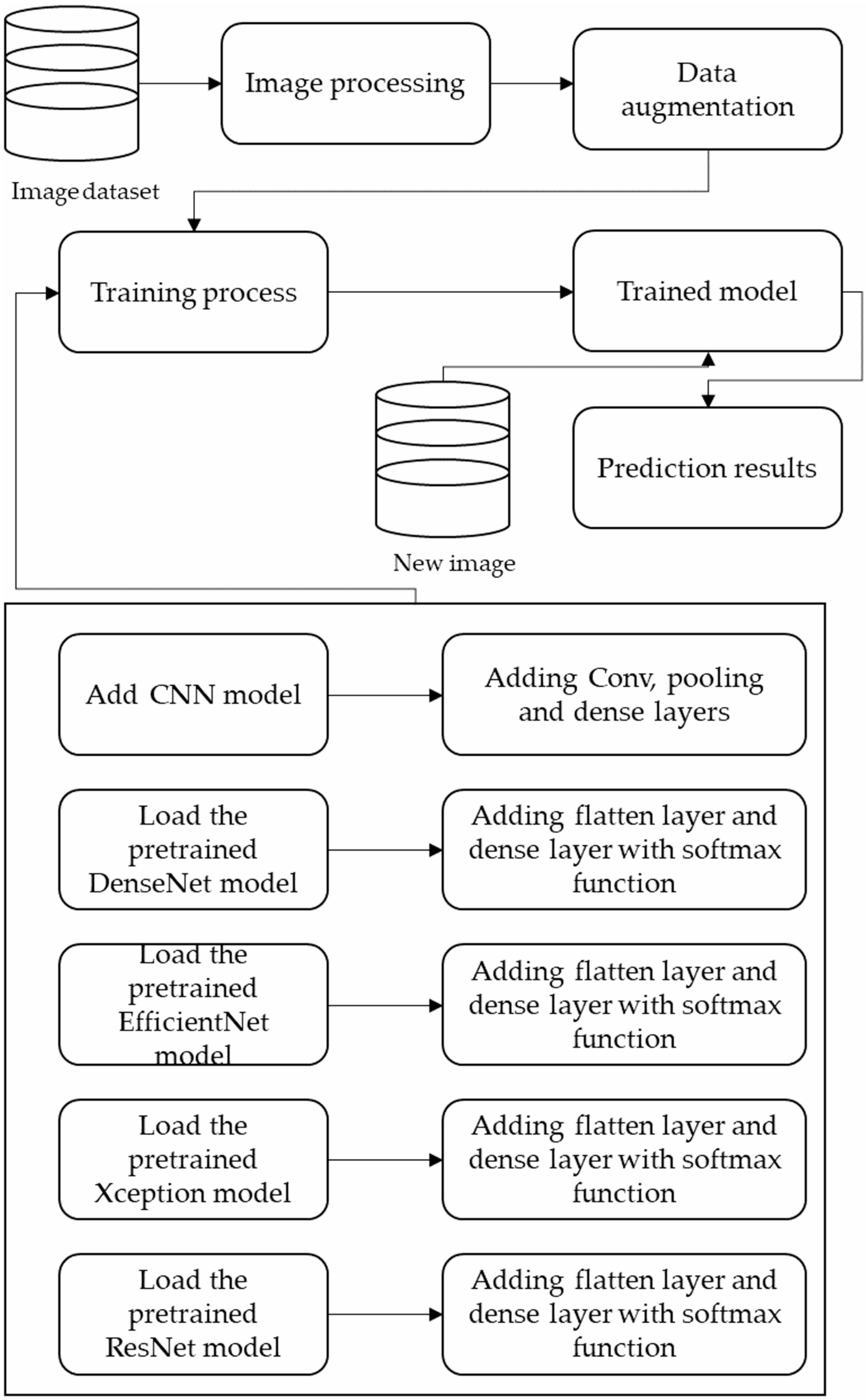

Fig. 1. The framework of this study.

XC uses depth-wise separable convolutions to work in spatial and depth dimensions simultaneously by separating these processes. Hence, it minimizes the number of parameters, computational cost, and preservation of symbolic power.

- Separable convolutions

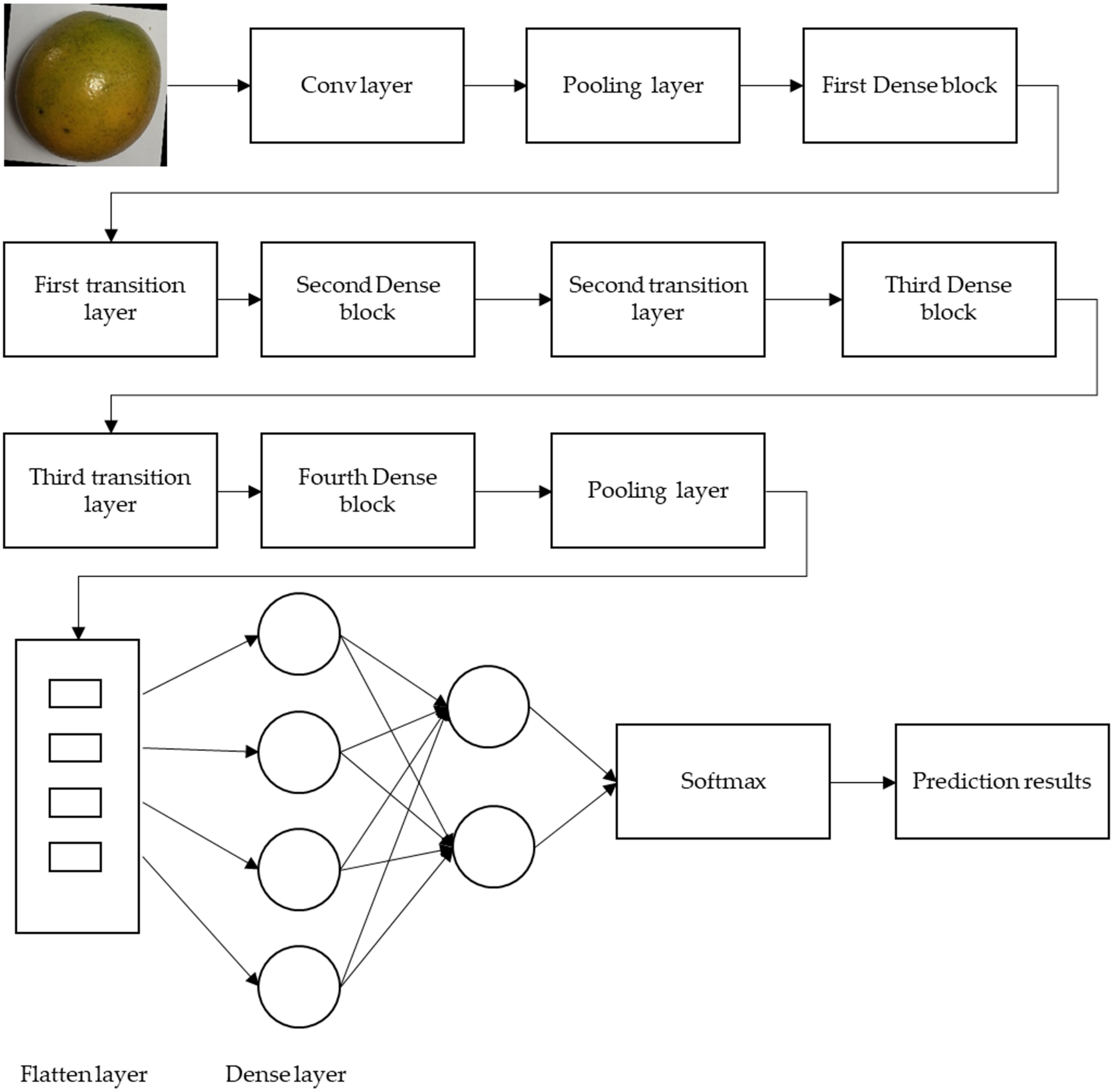

Fig. 2. Architecture of DenseNet121.

XC used a sequence of separable convolution blocks to improve the model's capability of capturing complex patterns in a smaller number of parameters. Every block has depth-wise separable convolution and linear transformation by pointwise convolution.

- Entry flow and exit flow

XC used the entry and exit flow to help in learning hierarchical presentations and facilitate the flow of information by the model. The entry flow is used to extract features, and the exit flow is used to refine these features to the outcome forecast.

- Skip connections

XC used skip connections to flow information through various layers and improve the efficient training model.

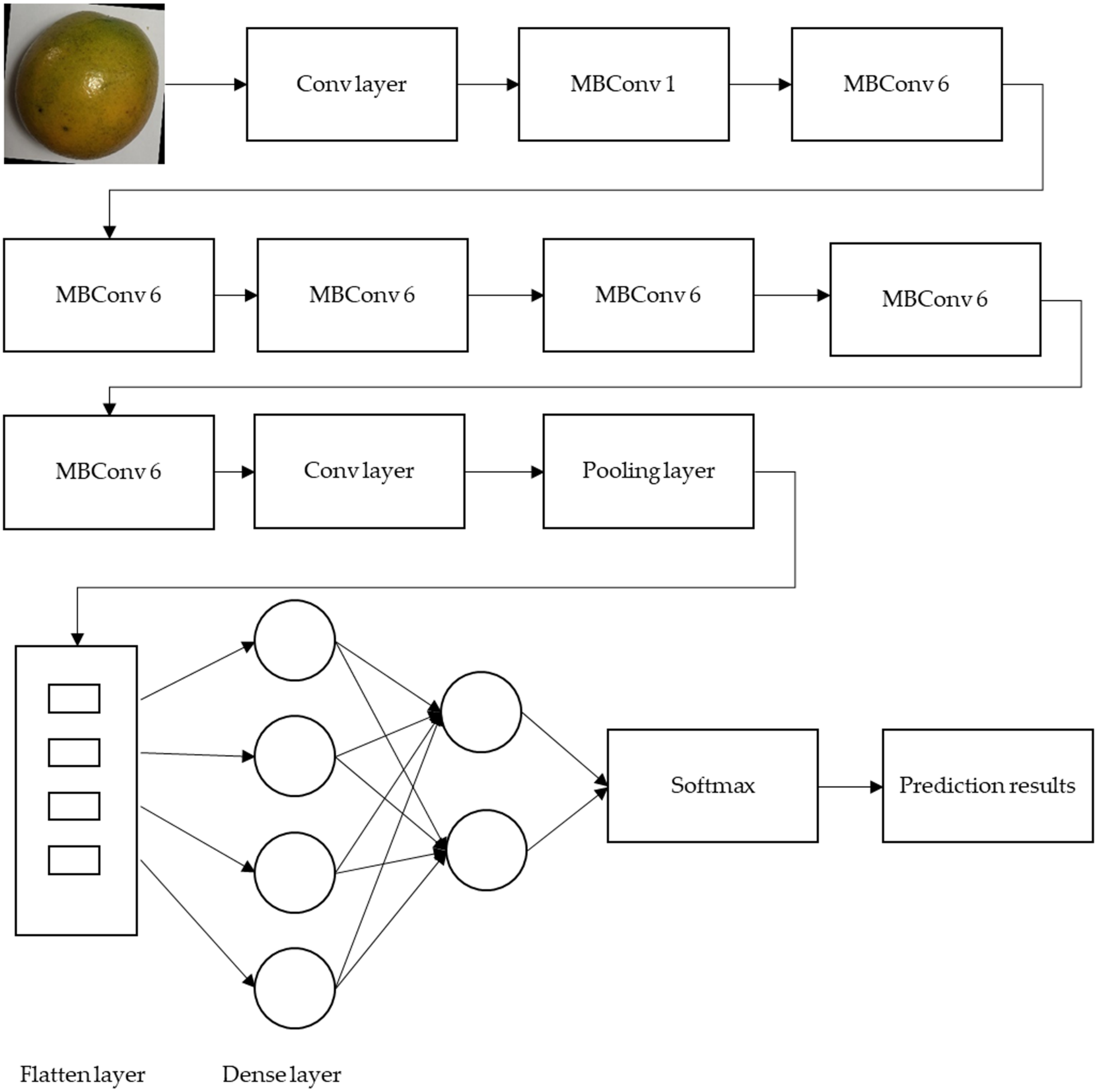

Fig. 3. The architecture of EfficientNet.

ResNet50 (RN)

RN refers to the residual network, which is a modification of ResNet. It has one average pooling layer, one max pooling layer, and 48 convolutional layers. There are 3 ConVs in every ConV and identification block. There are 23 million parameters that can be trained in the RN[53]. The architecture of RN is shown in Fig. 5.

The 50 in RN refers to the number of layers; it can be used for training large amounts of datasets to obtain better results and prediction outcomes. RN uses the residual connections to map the input to the desired outcome by learning a residual group. RN is used to overcome the vanishing gradient problems by learning much deeper structures. There are four main elements in RN such as:

- Convolutional layers

RN used serval ConV layers to extract features like shapes, textures, and edges from the input dataset. The ConV layers have various batch normalization and ReLU functions. Max pooling layers follow the ConV layers. Conv layers used a max pooling layer to minimize features' spatial dimension by preserving the essential features.

- Identity block

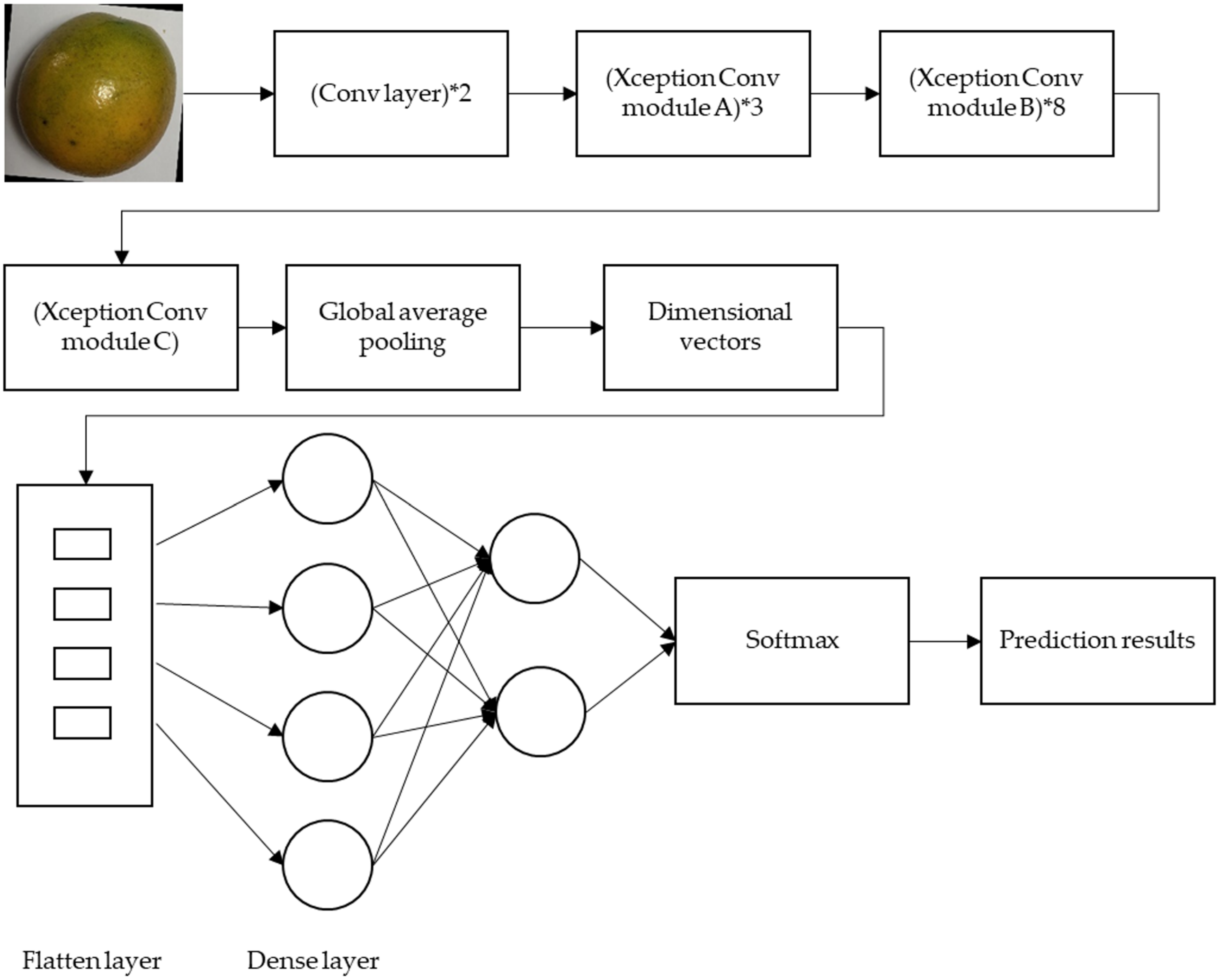

Fig. 4. Architecture of Xception.

It is used to add the input to the outcome and pass the input data into a sequence of convolutional layers. It also allows the model to learn residual functions to map the input to the outcome.

- Convolutional block

It is similar to the identity block. It uses ConV with 1*1 to minimize the number of filters before using ConV with 3*3.

- Fully connected layer

It uses a fully connected layer to produce output results and predictions. This layer used the softmax function to make the final multi-class classification.

Results and discussion
This section highlights the effectiveness of DL models in several case studies, demonstrating their potential to help farmers identify fruit plant diseases at an early stage. It also compares the performance of these models, showcasing the strengths of each one across different datasets. In addition, six case studies on fruit disease detection are introduced.

Case study 1 (Orange disease detection)
The first case study focuses on the detection of orange disease. Five DL models were used to assess their accuracy and performance in diagnosing this condition.

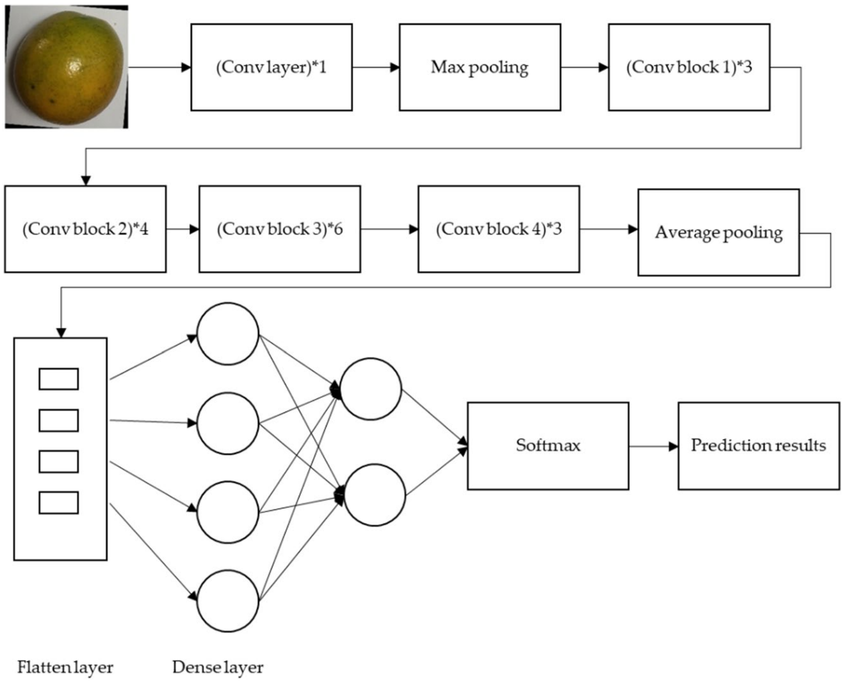

Fig. 5. Architecture of ResNet.

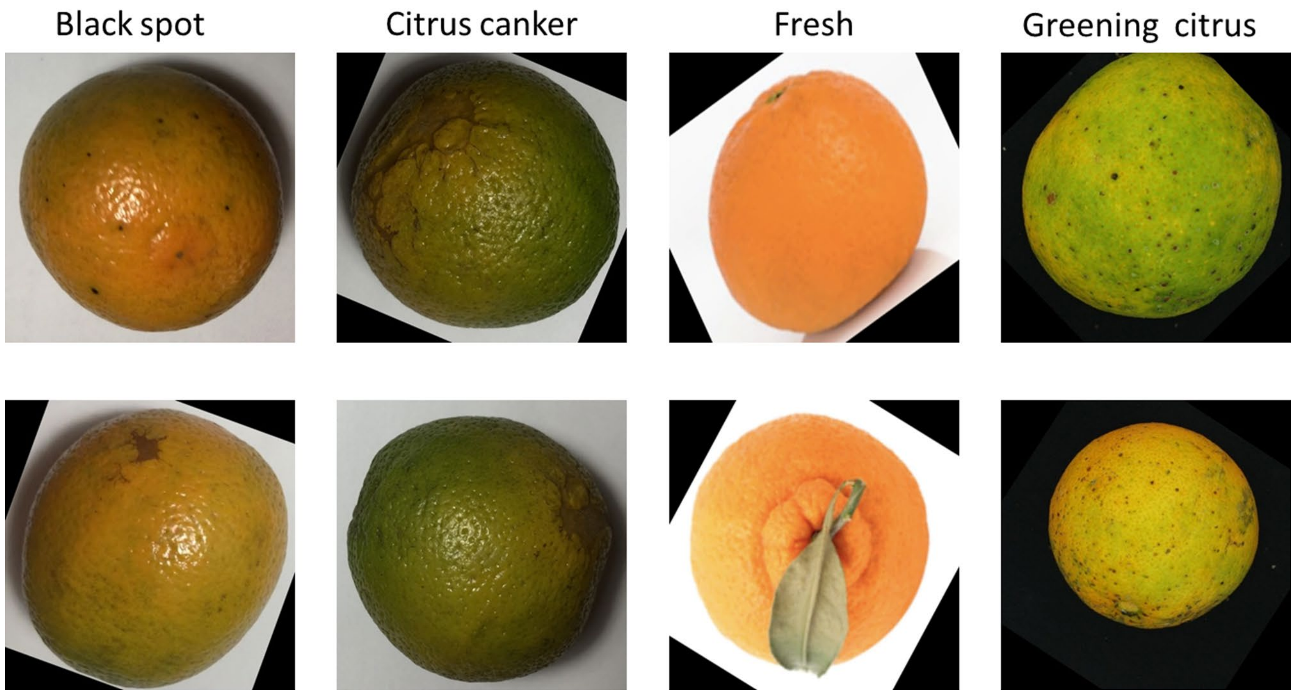

Fig. 6. Sample of orange disease dataset.

Dataset

This dataset was downloaded from[54]. It was developed to train DL models to classify orange disease. This dataset has four classes: fresh orange, citrus canker, black spot, and greening citrus. The size of this dataset is 145.59 MB. It has two folders: train and test data. Each folder has four classes. It has 184 images in the black spot class, 179 images in the citrus canker class, 281 images in the fresh class, and 347 images in the greening citrus class. Figure 6 shows a sample of the orange disease dataset.

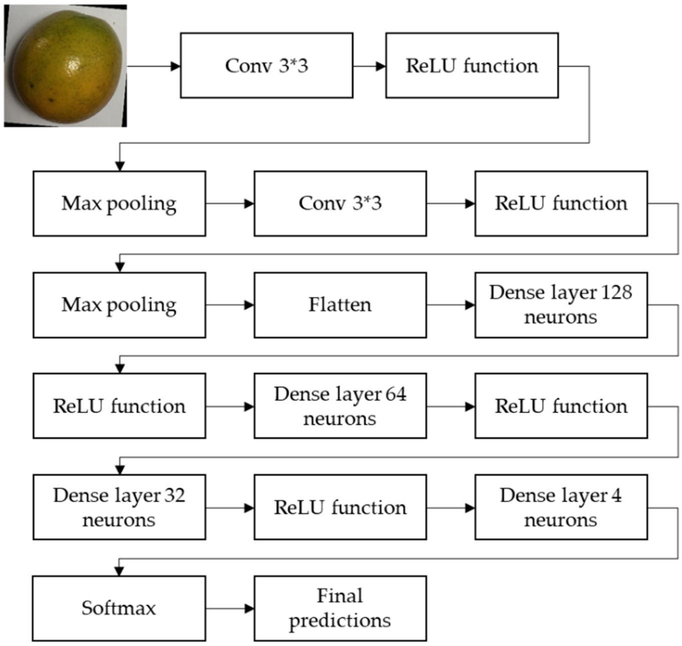

Fig. 7. The architecture of CNN algorithm.

	CNN	DN	EN	XC	RN
Optimizer (Opt)	**Adam**				
Learning rate (LR)	0.001				
Loss name	Sparse categorical cross entropy (SCCE)	SCCE	SCCE	SCCE	SCCE
Metrics	Accuracy (ACC)	ACC	ACC	ACC	ACC
Batch size	64	64	64	64	64
Epoch	10	10	10	10	10
Time per step	4 s/step	10 s/step	8 s/step	11 s/step	8 s/step
Epoch time	64s	144s	138s	190s	137s
Training time	612s	1,501	1,361	1,898s	1,363
Steps per epoch	16	16	16	16	16
ACC	100%	98.23%	99.10%	98.57%	100%
ACC validation	96.25%	93.9%	95.96%	92.93%	92.93%
Loss	0.0026	0.1546	0.0217	0.3307	0.0000877
Loss validation	0.0988	1.3670	0.2565	12.5598	1.0096

Table 1. Hyperparameters and results of DL models in orange disease.

In each folder, we applied IPP to the images. We resized the images into 256 and 256 with 3 channels. We put batch size with 64, label model is int, and labels are inferred. We show 991 images in four classes in the training process and 99 images in four classes in the testing process.

DL models

We applied the CNN model to this dataset to demonstrate its final predictions and superior performance. Figure 7 illustrates the architecture of the CNN algorithm used for fruit disease classification. Following this, we applied the other four deep learning (DL) models. Table 1 presents the hyperparameters and results for all five models. Figures 8, 9, 10 and 11, and 12 display the accuracy and loss metrics for each of the DL models. The results show that the CNN model achieves the highest accuracy and the shortest training time, followed by the EN model.

Figures 8, 9, 10, 11 and 12 present the loss and accuracy metrics for five DL models throughout the training and validation processes on the orange disease dataset. For the CNN model shown in Fig. 8, the training loss decreases consistently as the number of epochs increases, demonstrating that the model learns effectively during the training phase. The validation loss also shows a gradual decline but with slight fluctuations, indicating minor overfitting. In terms of accuracy, both training and validation accuracy improve steadily, with validation accuracy converging close to the training accuracy by the end of the epochs. This suggests that the CNN model

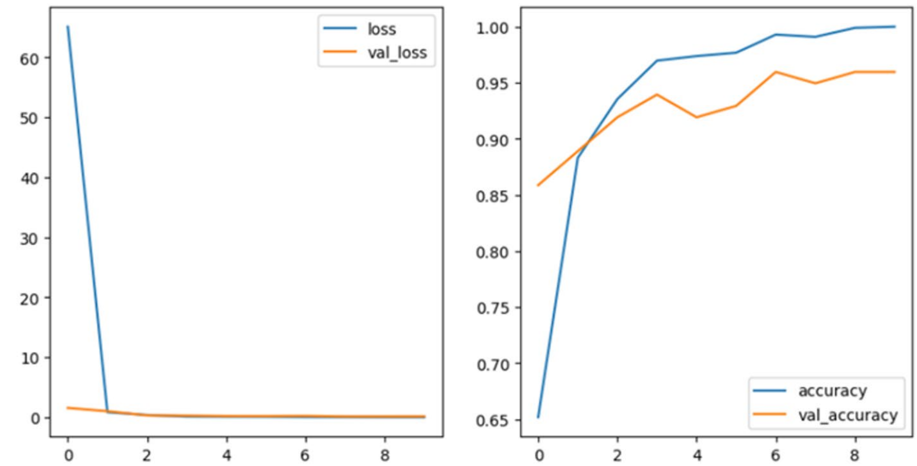

Fig. 8. Loss and accuracy for orange disease dataset by the CNN algorithm.

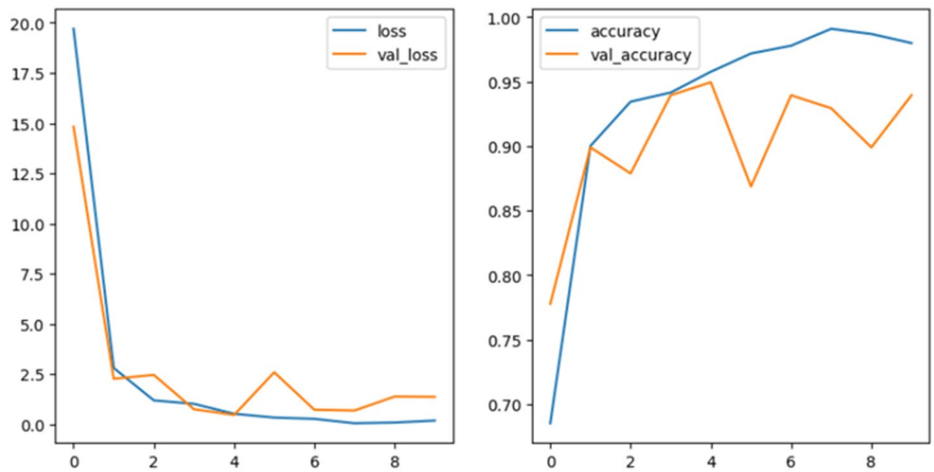

Fig. 9. Loss and accuracy for orange disease dataset by the DN algorithm.

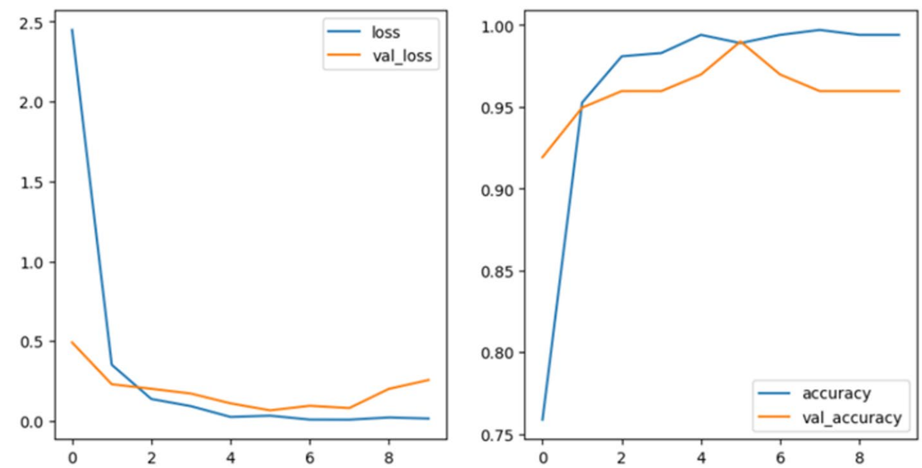

Fig. 10. Loss and accuracy for orange disease dataset by the EN algorithm.

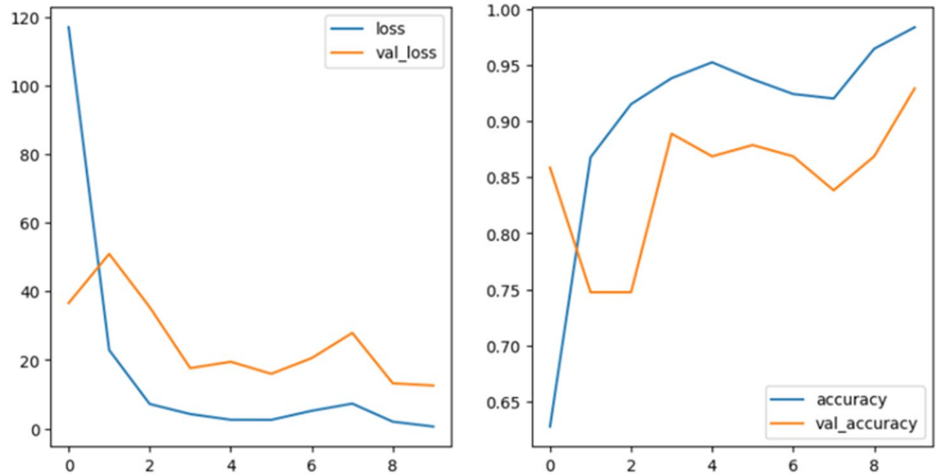

Fig. 11. Loss and accuracy for orange disease dataset by the XC algorithm.

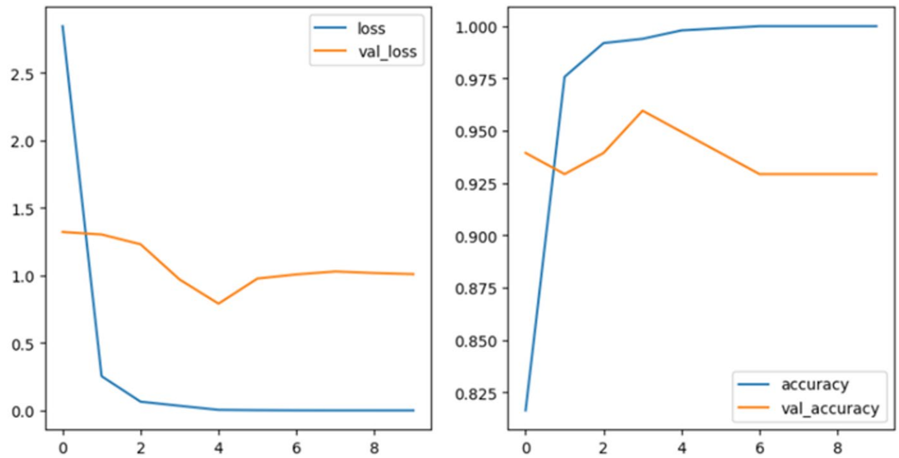

Fig. 12. Loss and accuracy for orange disease dataset by the RN algorithm.

achieves a balance between learning and generalizing well to unseen data. The DenseNet121 model exhibits a sharp decline in training loss during the initial epochs, followed by a plateau, indicating efficient convergence, as shown in Fig. 9. The validation loss mirrors this trend but has small spikes, suggesting occasional challenges in generalizing to new data. The accuracy graphs show a clear improvement for both the training and validation phases, with the validation accuracy remaining slightly lower than the training accuracy, reflecting good but not perfect generalization. The EfficientNet model, in Fig. 10, shows a smooth and rapid reduction in training loss, with validation loss following a similar trajectory. However, the validation loss shows minor inconsistencies, likely due to the model adapting to complex features in the dataset. The accuracy graphs indicate that the EfficientNet model achieves high training and validation accuracy, with the validation accuracy approaching the training accuracy. This suggests that EfficientNet efficiently handles the fruit disease dataset. The Xception model in Fig. 11 shows the training loss decreases steadily, while the validation loss shows some fluctuations, indicating sensitivity to variations in the validation dataset. Despite this, both training and validation accuracy improve consistently across epochs. The validation accuracy lags slightly behind the training accuracy, demonstrating the model's ability to learn complex patterns while maintaining reasonable generalization. The ResNet50 model achieves a rapid decrease in training loss during the early epochs, followed by a slower decline as the model converges, as shown in Fig. 12. The validation loss initially decreases but exhibits some spikes, reflecting occasional difficulties in generalization. The accuracy curves show steady improvements, with training accuracy increasing consistently and validation accuracy following closely. This indicates that ResNet50 effectively balances learning and generalization, making it a strong candidate for fruit disease detection.

Case study 2 (Grape disease detection)

This part shows the results of DL models in grape disease.

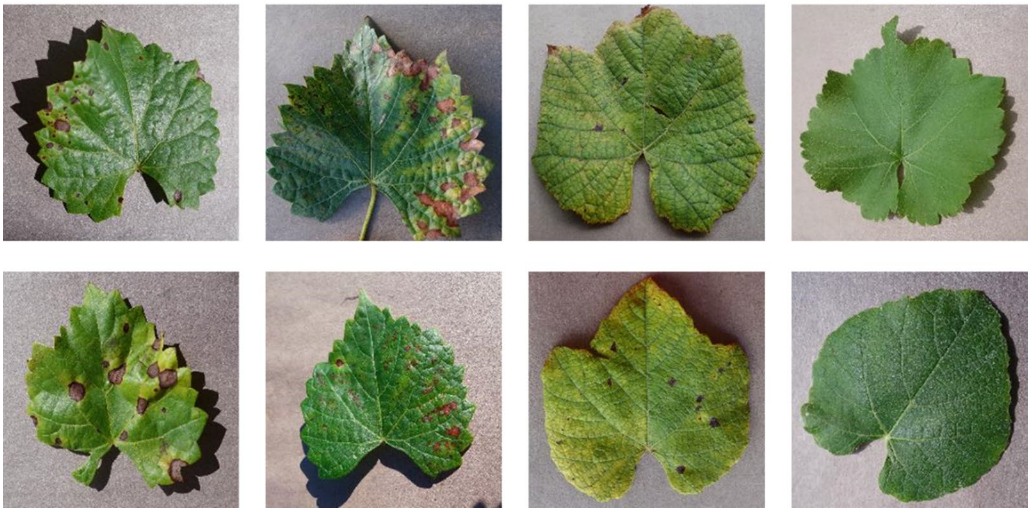

Fig. 13. Sample of grape disease dataset.

	CNN	DN	EN	XC	RN
Opt	**Adam**				
LR	0.001				
Loss name	SCCE	SCCE	SCCE	SCCE	SCCE
Metrics	ACC	ACC	ACC	ACC	ACC
Batch size	64	64	64	64	64
Epoch	10	10	10	10	10
Time per step	3 s/step	8 s/step	7 s/step	11 s/step	7 s/step
Epoch time	138 s	362 s	352 s	473 s	368
Training time	1381 s	3618 s	3519 s	4734 s	3686 s
Steps per epoch	45	45	45	45	45
ACC	98.235	99.19%	98.85%	96.23%	100%
ACC validation	92.36%	96.55%	99.14%	83.48%	98.15%
Loss	0.0513	0.0299	0.0334	0.5262	0.00037
Loss validation	0.2630	0.1630	0.0484	6.1152	0.1556

Table 2. Hyperparameters and results of DL models in grape disease.

Dataset
This dataset was downloaded from[55]. It was developed to train DL models to classify grape disease. This dataset has four classes: black rot, esca (black measles), leaf blight, and healthy. The size of this dataset is 72.68 MB. It has one folder. Each folder has four classes. It has 1180 images in the black rot class, 1383 images in the esca (black measles) class, 1076 images in leaf blight, and 423 images in the healthy class. Figure 13 shows a sample of the grape disease dataset We divided images into training and validation data. The train data has 2843 images and the validation data has 811 images. In each folder, we applied IPP to the images. We resized the images into 224 and 224 with 3 channels. We put the batch size at 64, the label model is int, and the labels are inferred.

DL models
We applied the CNN model to this dataset to demonstrate its final predictions and superior performance. The architecture of the CNN model used for orange disease detection was adapted for this study. Next, we applied the other four deep learning (DL) models. Table 2 presents the hyperparameters and results for all five models. Figures 14, 15, 16 and 17, and 18 display the accuracy and loss metrics for each DL model.

Figures 14, 15, 16, 17 and 18 illustrate the loss and accuracy plots for five DL models throughout the training and validation processes on the grape disease dataset. The CNN model, in Fig. 14, shows the training loss decreases sharply during the initial epochs, indicating that the model learns rapidly. The validation loss also follows a similar downward trend but stabilizes with minor fluctuations. Training accuracy improves steadily and reaches a high value by the end of training. Validation accuracy increases more gradually, eventually converging close to 90%. This indicates that the CNN model effectively generalizes to unseen data, though

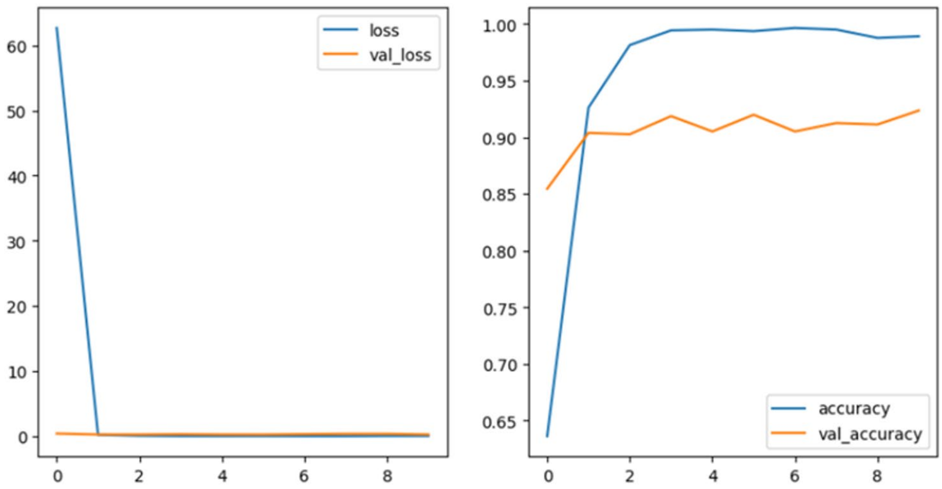

Fig. 14. Loss and accuracy for grape disease dataset by the CNN algorithm.

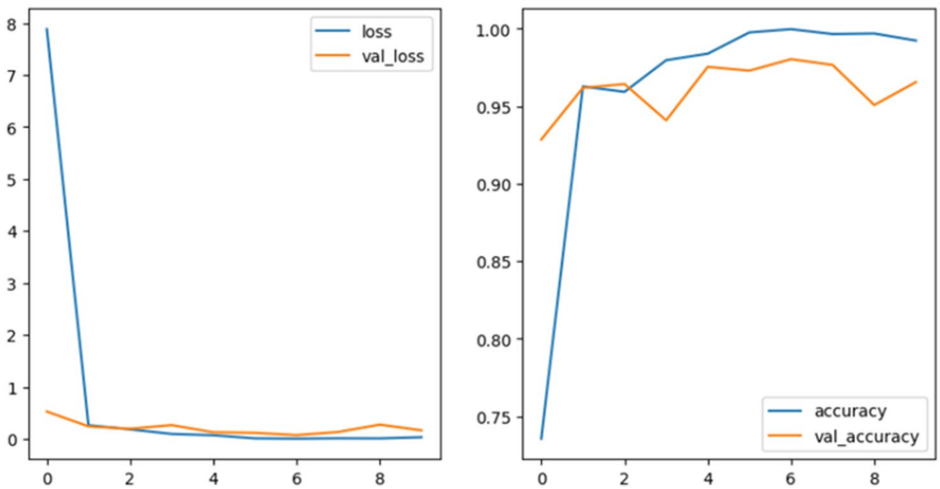

Fig. 15. Loss and accuracy for grape disease dataset by the DN algorithm.

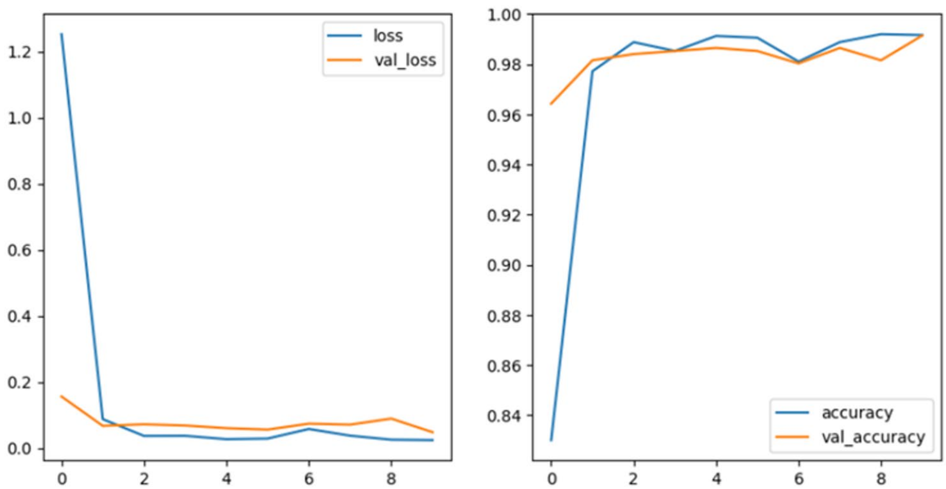

Fig. 16. Loss and accuracy for grape disease dataset by the EN algorithm.

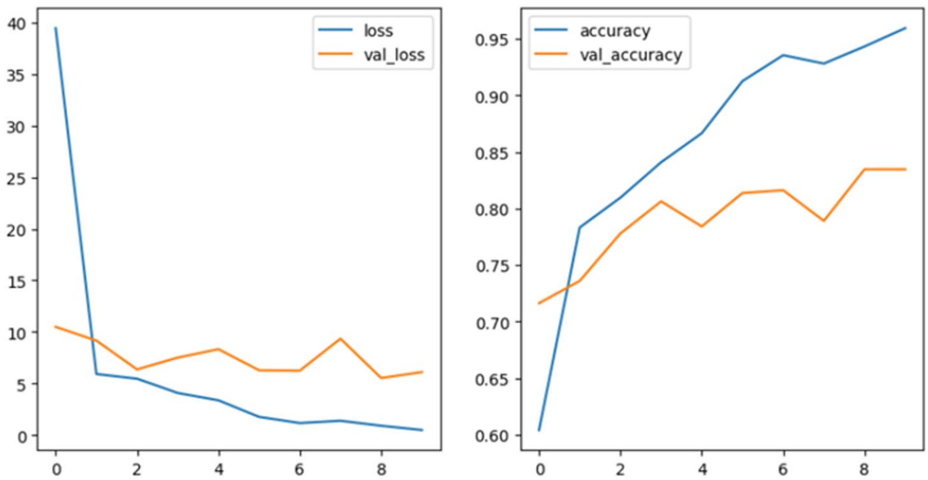

Fig. 17. Loss and accuracy for grape disease dataset by the XC algorithm.

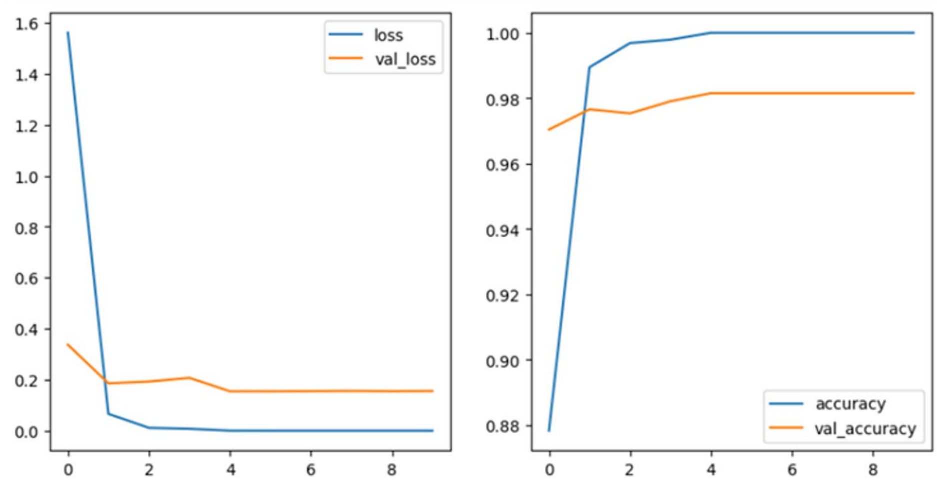

Fig. 18. Loss and accuracy for grape disease dataset by the RN algorithm.

with slight signs of overfitting. The training loss drops consistently in the DenseNet model shown in Fig. 15, indicating strong learning progress. The validation loss exhibits occasional fluctuations but generally decreases, showing that the model performs well on unseen data while occasionally struggling with complex patterns. Both training and validation accuracy improve significantly during the training process. Validation accuracy closely follows training accuracy, indicating good generalization and effective performance in detecting fruit diseases. The EfficientNet mode, in Fig. 16, shows the training loss decreases quickly during the early epochs, stabilizing towards the end. Validation loss exhibits small but noticeable fluctuations, suggesting that the model captures intricate patterns in the data while adapting to the dataset's variability. The training accuracy of the model rises steadily, while validation accuracy follows a similar trend with a slight lag. This indicates that EfficientNet maintains a good balance between learning and generalizing, making it a strong performer for this task. The Xception model evaluation, in Fig. 17, shows the training loss's sharp initial decline, stabilizing as the model converges. The validation loss exhibits more significant fluctuations compared to other models, highlighting sensitivity to the validation dataset's complexity. The model's training accuracy improves steadily and reaches a high value. Validation accuracy, while improving, is slightly lower than training accuracy due to the aforementioned fluctuations. The Xception model is effective but requires fine-tuning to handle validation data variations better. Figure 18 shows the ResNet50 model performance, and it can be seen that the training loss decreases rapidly in the early epochs, followed by a gradual decline. Validation loss is relatively stable with minimal fluctuations, indicating that the model generalizes well without significant overfitting. The model's training accuracy improves consistently, closely following validation accuracy.

Fig. 19. Sample of mango disease dataset.

	CNN	DN	EN	XC	RN
Opt	**Adam**				
LR	0.001				
Loss name	CCE	CCE	CCE	CCE	CCE
Metrics	ACC	ACC	ACC	ACC	ACC
Batch size	40	40	40	40	40
Epoch	10	10	10	10	10
Time per step	2 s/step	4 s/step	4 s/step	5 s/step	4 s/step
Epoch time	170 s	316 s	294 s	371	317 s
Training time	1708 s	3168 s	2941 s	3719 s	3170 s
Steps per epoch	70	70	70	70	70
ACC	73.84%	91.17%	94.07%	60.93%	96.85%
ACC validation	76.83%	94.50%	95.83%	70.33%	96.17%
Loss	0.7296	1.6021	0.6140	50.52	0.5047
Loss validation	0.6078	0.8705	0.5931	30.7733	0.969

Table 3. Hyperparameters and results of DL models in Mango disease.

Case study 3 (Mango disease detection)

Dataset

This dataset was downloaded from[56] and is designed to train deep learning (DL) models to classify mango diseases. It contains eight classes: Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Healthy, Powdery Mildew, and Sooty Mould. The dataset size is 140.67 MB and consists of a single folder with eight subfolders, each representing one disease class. There are 500 images in each class, for a total of 4000 images. Figure 19 shows a sample from the mango disease dataset.

For image preprocessing, we applied Image Processing Procedures (IPP) to all images. We resized the images to 224×224 pixels with three color channels (RGB). A batch size of 40 was used during training, with 2800 images (350 images per class) in the training set and 600 images (75 images per class) in the test set. The class mode was set to "categorical," the color mode to "RGB," and the data was shuffled for training.

DL models

We applied the CNN model in this dataset to show the final prediction and better performance. We used the CNN model as in the orange disease classification. Then we apply the other four DL models. Table 3 shows the hyperparameters and results of DL models. Figures 20, 21, 22, 23 and 24 show the accuracy and loss for five DL models.

Figures 20, 21, 22, 23 and 24 illustrate the loss and accuracy plots for five DL models throughout the training and validation processes on the mango disease dataset. The CNN model, shown in Fig. 20, exhibits a rapid initial decrease in training loss, indicating effective learning in early epochs. However, the loss curve plateaus shortly after, suggesting potential overfitting. The accuracy curve shows a steady increase, reaching a plateau as well. This indicates that further training might not improve the model's performance significantly. The fluctuations

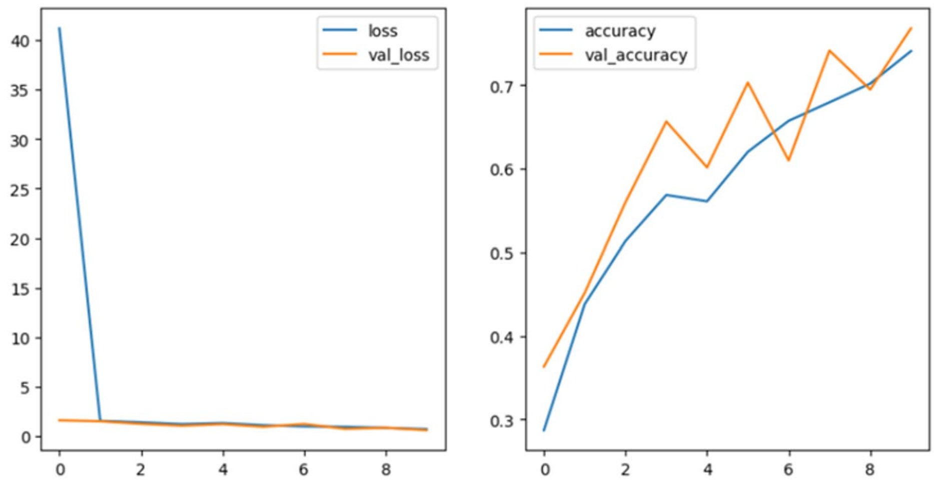

Fig. 20. Loss and accuracy for mango disease dataset by the CNN algorithm.

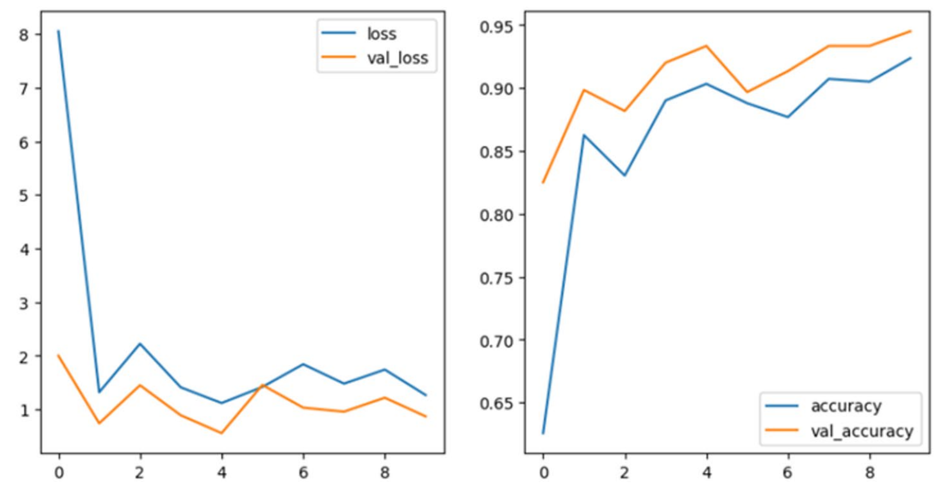

Fig. 21. Loss and accuracy for mango disease dataset by the DN algorithm.

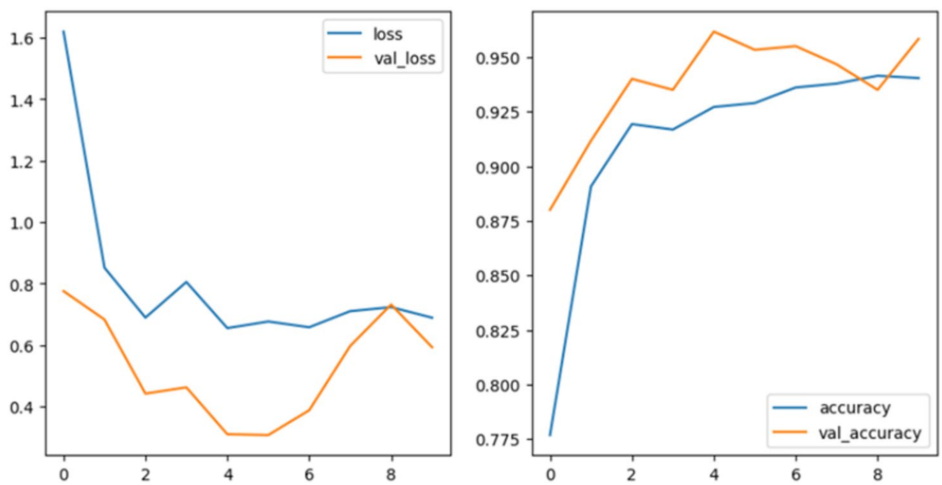

Fig. 22. Loss and accuracy for mango disease dataset by the EN algorithm.

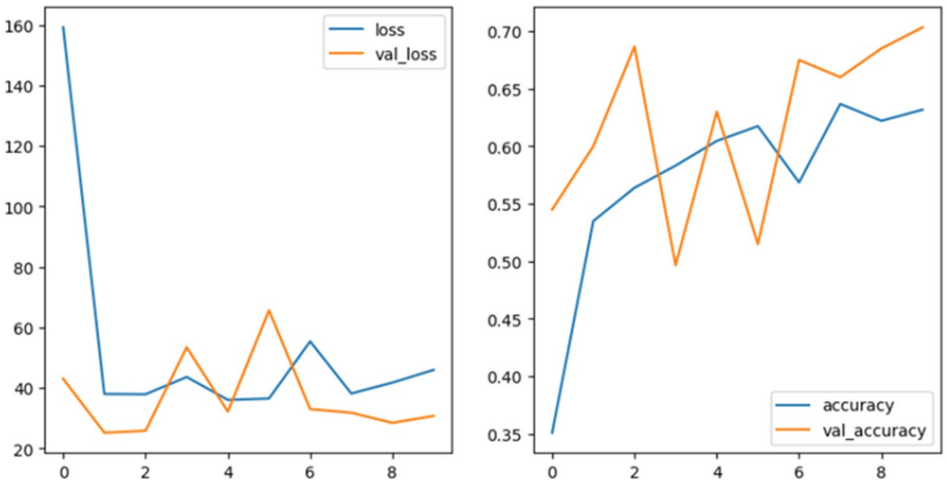

Fig. 23. Loss and accuracy for mango disease dataset by the XC algorithm.

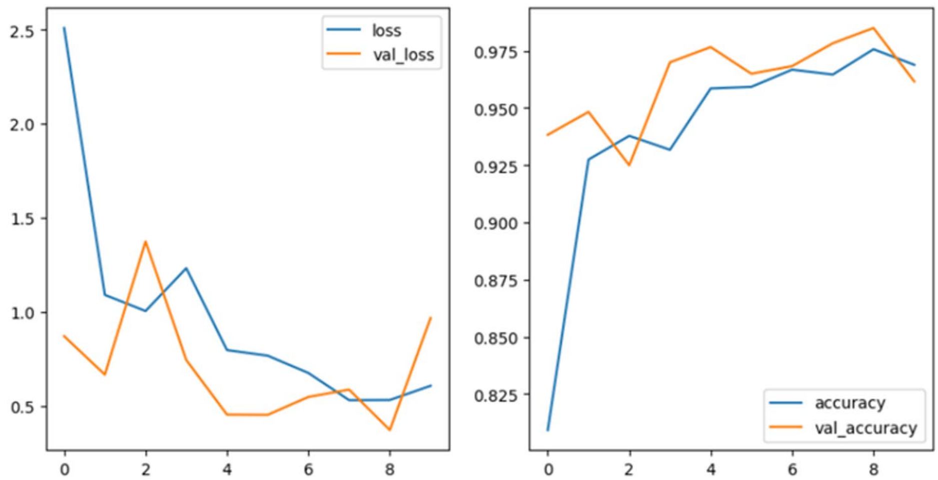

Fig. 24. Loss and accuracy for mango disease dataset by the RN algorithm.

in the validation curve indicate the variations in the validation dataset that it is less familiar with than the training data. DenseNet in Fig. 21 follows a similar trend to the CNN model. It demonstrates a rapid initial decrease in loss, followed by a plateau. However, DenseNet achieves a slightly lower final loss compared to CNN, suggesting better generalization. The accuracy curve for DenseNet also shows a steady increase, reaching a plateau. It achieves a slightly higher final accuracy compared to CNN. EfficientNet's loss curve, in Fig. 22, shows a more gradual decrease compared to the previous models, potentially indicating a slower learning process but potentially better generalization. The loss curve plateaus after a few epochs, similar to other models. The model's validation accuracy is higher than the training accuracy, indicating that the validation dataset may contain simpler or less diverse examples compared to the training dataset, making it easier for the model to classify correctly. Figures 23 and 24 show the Xception and the ResNet50 model performance curves, illustrating the variation in the validation accuracy, which indicates a slight overfitting to the training data.

Case study 4 (Banana disease detection)

Dataset

This dataset was downloaded from[57]. It was developed to train DL models to classify banana disease. This dataset has four classes: cordana, healthy, pestalotiopsis, and sigatoka. The size of this dataset is 40.05 MB. It has one folder. Each folder has four classes. It has 400 images in each class. It has 1280 images in train data and 320 images in validation data. Figure 25 shows a sample of the banana disease dataset. In each folder, we applied IPP in images. We resized the images into 224 and 224 with 3 channels. We put batch size with 32, rescale images and class mode is categorical.

Fig. 25. Sample of the banana dataset.

	CNN	DN	EN	XC	RN
Opt	Adam				
LR	0.001				
Loss name	SCCE	CCE	CCE	CCE	CCE
Metrics	ACC	ACC	ACC	ACC	ACC
Batch size	32	64	64	64	64
Epoch	5	5	5	5	5
Time per step	6 s/step	7 s/step	6 s/step	8 s/step	6 s/step
Epoch time	256 s	136 s	125 s	168 s	123 s
Training time	1280 s	680 s	627 s	844 s	617 s
Steps per epoch	40	20	20	20	20
ACC	89.66%	100%	26%	100%	49.76%
ACC validation	69.69%	94.06%	25%	92.50%	46.88%
Loss	0.3230	0.0015	2.57	0.00785	1.1584
Loss validation	1.0220	0.1806	1.8659	0.3820	1.2205

Table 4. Hyperparameters and results of DL models in banana disease.

DL models

We applied the CNN model in this dataset to show the final prediction and better performance. We used the CNN model as in orange disease. Then we apply the other four DL models. Table 4 shows the hyperparameters and results of DL models. Figures 26, 27, 28, 29 and 30 show the accuracy and loss for five DL models.

Figures 26, 27, 28, 29 and 30 illustrate the loss and accuracy plots for five DL models throughout the training and validation processes on the banana disease dataset. Figure 26 illustrates the training and validation performance of the CNN model. The training loss exhibits a consistent downward trend, suggesting effective learning over the epochs. The validation loss also decreases initially but stabilizes with minor fluctuations, indicating some sensitivity to the complexity of the validation dataset. Both training and validation accuracy curves show a steady increase, with the validation accuracy closely following the training accuracy curve. This suggests that the CNN model generalizes well, without significant overfitting or underfitting. Figure 27 presents the training and validation performance of the DenseNet model. The training loss decreases smoothly, demonstrating effective error minimization during learning. However, the validation loss shows some oscillations, indicating potential difficulties in consistent performance on unseen data. Training accuracy improves steadily and reaches a high value, while validation accuracy shows noticeable fluctuations but a general upward trend. This suggests strong learning, although the DenseNet model might benefit from further fine-tuning to stabilize its validation performance. Figure 28 displays the training and validation performance of the EfficientNet model. The EfficientNet model demonstrates a rapid decline in training loss, indicating efficient learning. The validation loss decreases initially but exhibits occasional fluctuations, potentially due to the complexity of the validation dataset. Training accuracy increases consistently, while validation accuracy also improves but with slight instability. This suggests that EfficientNet performs well but might be sensitive to specific variations in the validation dataset. Figure 29 illustrates the training and validation performance of the Xception model. The

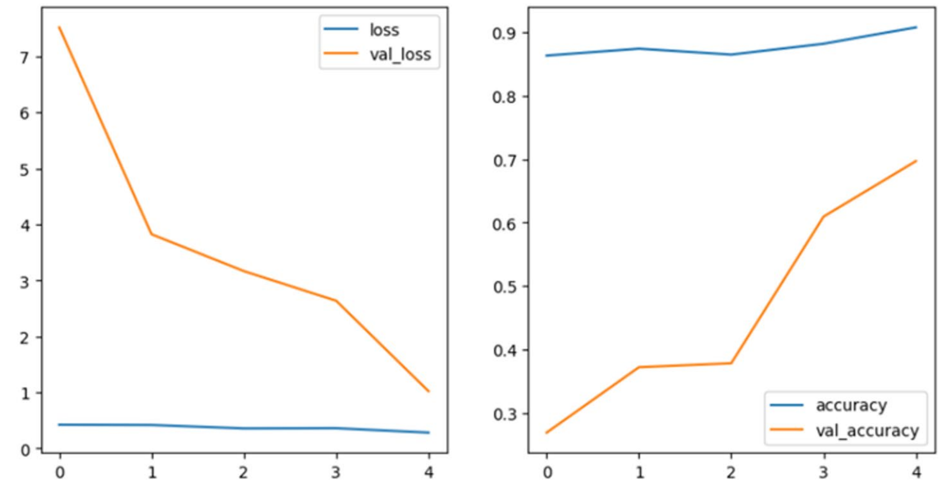

Fig. 26. Loss and accuracy for banana disease dataset by the CNN algorithm.

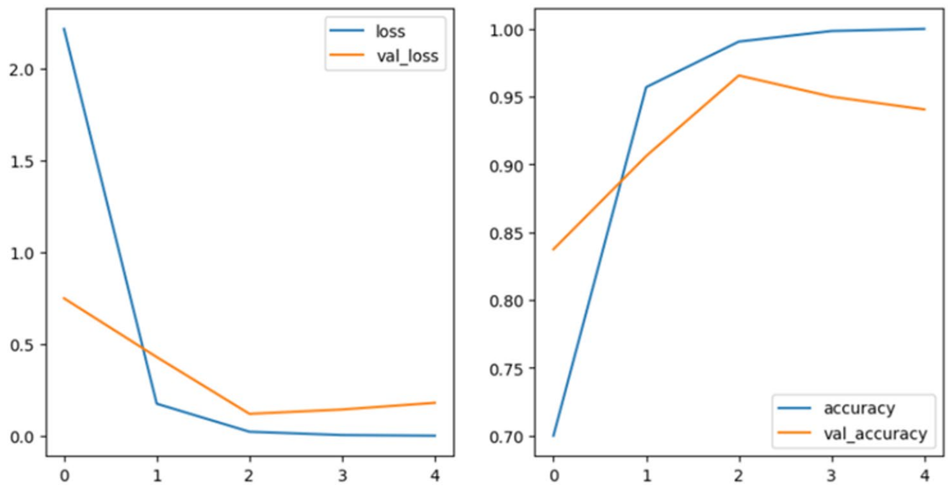

Fig. 27. Loss and accuracy for banana disease dataset by the DN algorithm.

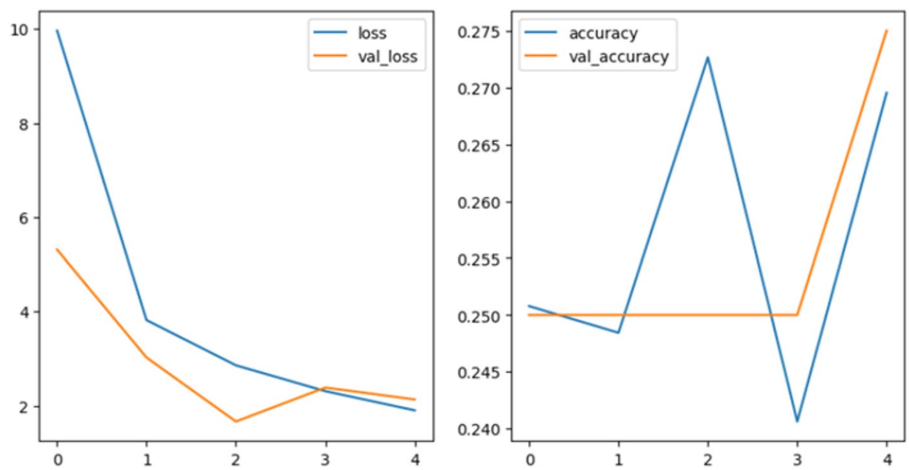

Fig. 28. Loss and accuracy for banana disease dataset by the EN algorithm.

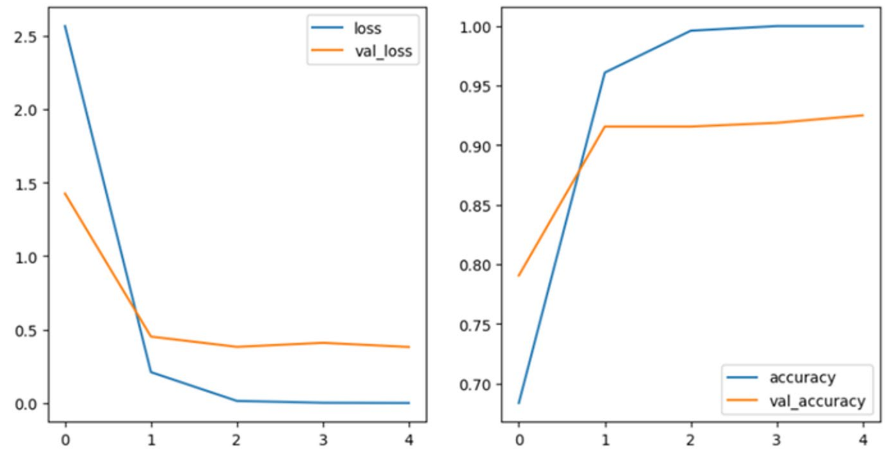

Fig. 29. Loss and accuracy for banana disease dataset by the XC algorithm.

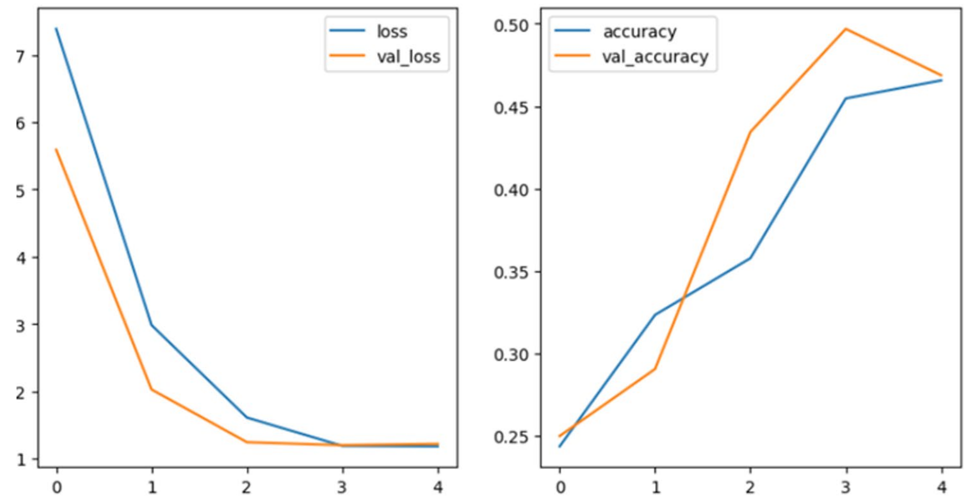

Fig. 30. Loss and accuracy for banana disease dataset by the RN algorithm.

training loss decreases sharply during the initial epochs, reflecting effective optimization. The training accuracy curve rises steadily, while validation accuracy follows the training accuracy before stabilizing at a high value. This behavior indicates that the Xception model captures features effectively but might need additional adjustments to handle data variability more consistently. Figure 30 presents the training and validation performance of the ResNet50 model. The training loss curve decreases steadily, highlighting consistent learning by the ResNet50 model. The validation loss also declines but remains relatively stable, with minimal fluctuations, indicating strong generalization capability. Both training and validation accuracy curves increase and converge closely, with validation accuracy slightly surpassing training accuracy at some points.

Case study 5 (Guava disease detection)

Dataset

This dataset was downloaded from[58]. It was developed to train DL models to classify guava disease. This dataset has five classes Disease Free, Phytophthora, Red Rust, Scab, and Styler and Root. The size of this dataset is 312.33 MB. It has two folders: train and test data. Each folder has five classes. It has 802 images in Disease Free, 845 images in Phytophthora, 993 images in Red Rust, 689 images in Scab, and 923 images in Styler and Root. Figure 31 shows a sample of the guava disease dataset. In each folder, we applied IPP to the images. We resized the images into 224 and 224 with 3 channels. We put the batch size at 64, the label model is int, and the labels are inferred. We show 4252 images in five classes in the training process and 1068 images in five classes in the testing process.

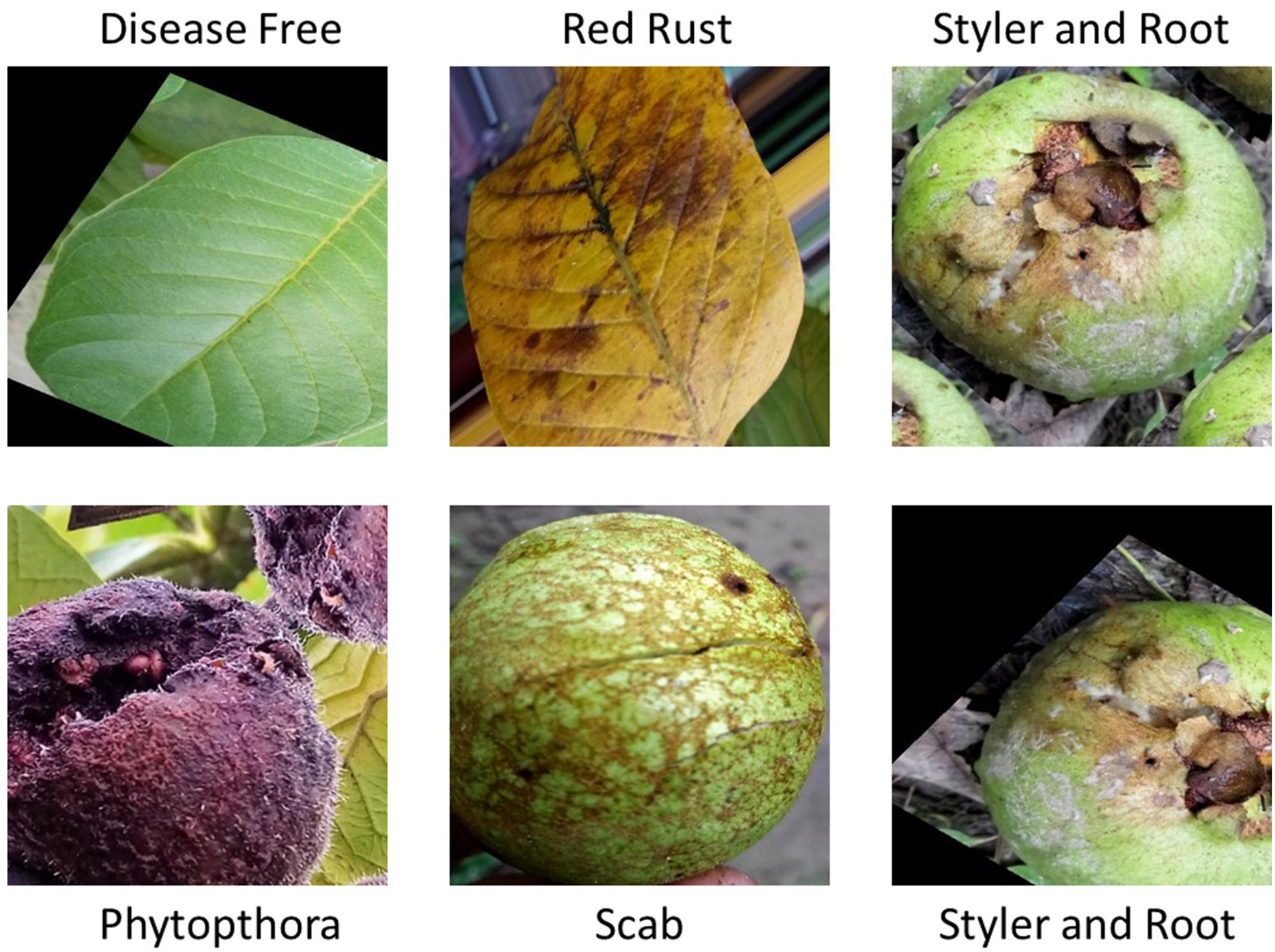

Fig. 31. Sample of guava disease dataset.

	CNN	DN	EN	XC	RN
Opt	**Adam**				
LR	0.001				
Loss name	SCCE	SCCE	SCCE	SCCE	SCCE
Metrics	ACC	ACC	ACC	ACC	ACC
Batch size	64	64	64	64	64
Epoch	10	5	5	5	5
Time per step	3 s/step	7 s/step	6 s/step	8 s/step	6 s/step
Epoch time	193 s	475	416 s	559 s	433 s
Training time	1931 s	2326 s	2082 s	2795	2165 s
Steps per epoch	67	67	67	67	67
ACC	98.97%	97.44%	98.83%	90.27%	99.50%
ACC validation	53.28%	93.35%	96.16%	75.37%	96.72%
Loss	0.0534	0.2045	0.0560	4.2534	0.0375
Loss validation	2.999	0.7688	0.2913	22.7974	0.6200

Table 5. Hyperparameters and results of DL models in guava disease.

DL models

We applied the CNN model in this dataset to show the final prediction and better performance. We used the CNN model as in orange disease. Then, we apply the other four DL models. Table 5 shows the hyperparameters and results of DL models. Figures 32, 33, 34, 35 and 36 show the accuracy and loss for five DL models.

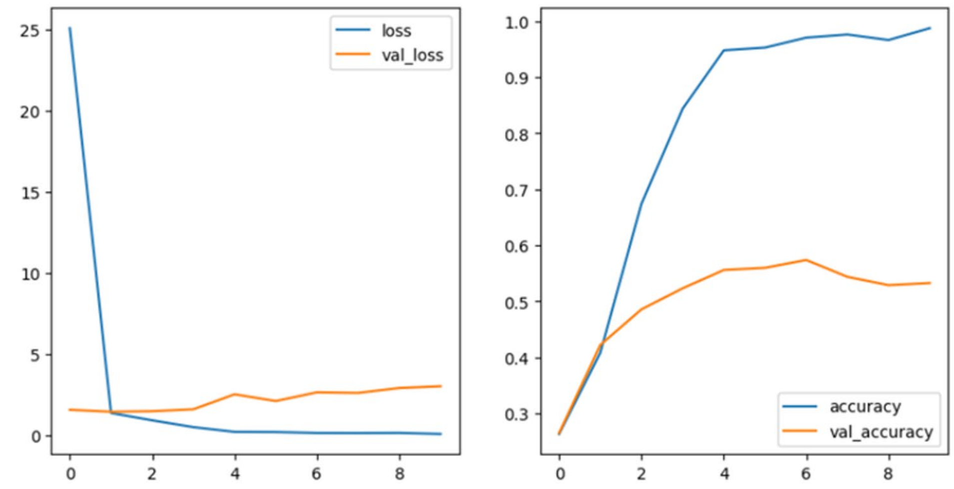

Fig. 32. Loss and accuracy for guava disease dataset by the CNN algorithm.

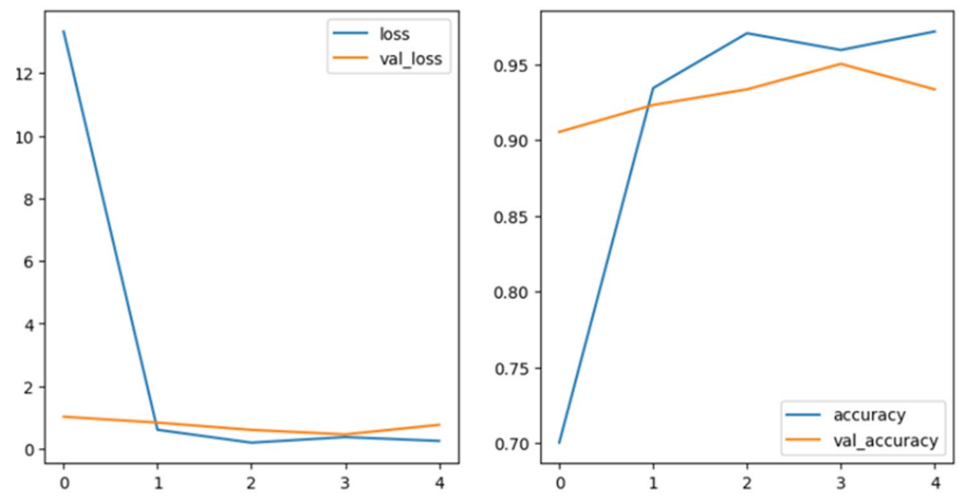

Fig. 33. Loss and accuracy for guava disease dataset by the DN algorithm.

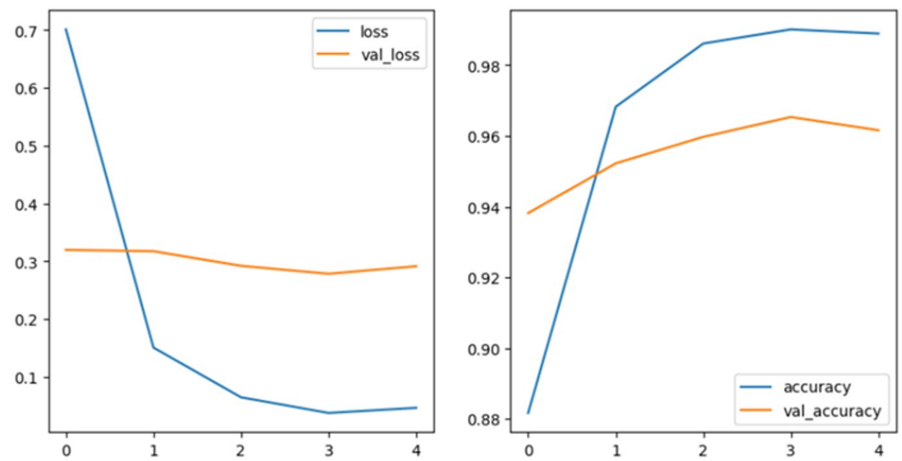

Fig. 34. Loss and accuracy for guava disease dataset by the EN algorithm.

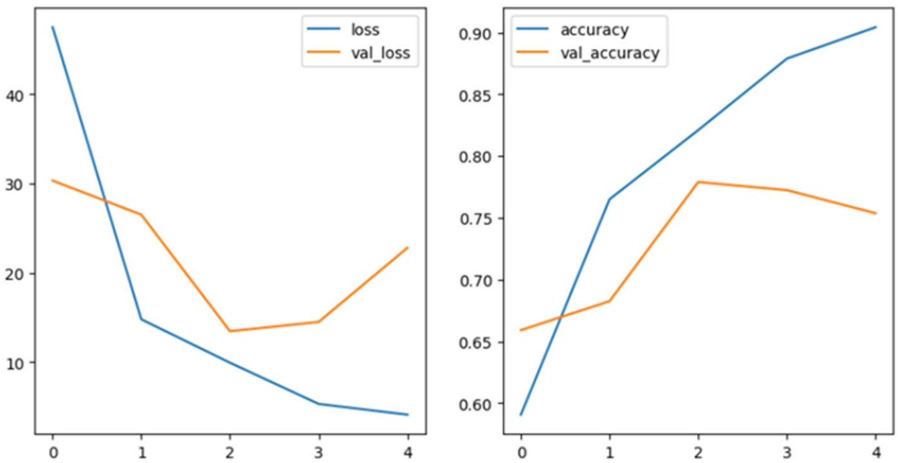

Fig. 35. Loss and accuracy for guava disease dataset by the XC algorithm.

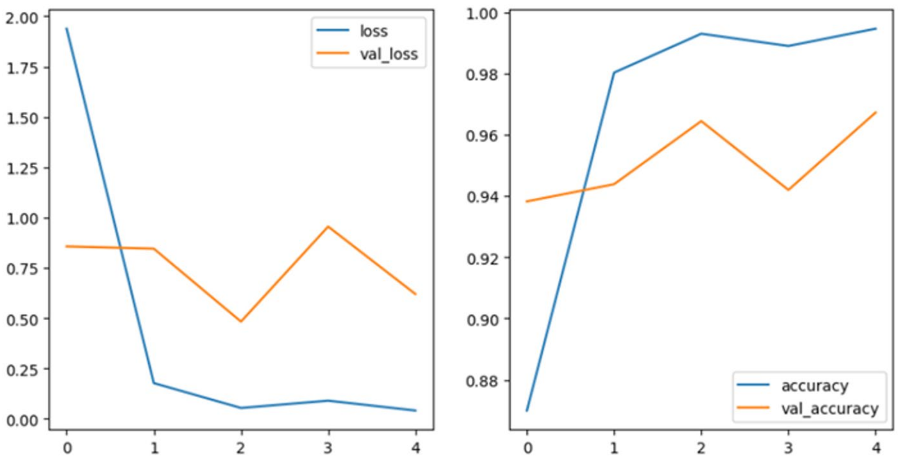

Fig. 36. Loss and accuracy for guava disease dataset by the RN algorithm.

Figures 32, 33, 34, 35 and 36 illustrate the loss and accuracy plots for five DL models throughout the training and validation processes on the guava disease dataset. Figure 32 illustrates the training and validation performance of the CNN model. The validation loss is higher than the training, indicating that the model performs better on the training data than the unseen validation data. The validation accuracy is lower than the training accuracy, showing that the model struggles to generalize well to unseen data. Figure 33 presents the training and validation performance of the DenseNet model. The training loss decreases smoothly, demonstrating effective error minimization during learning. Training accuracy improves steadily and reaches a high value, while validation accuracy shows noticeable fluctuations but a general upward trend. This suggests strong learning, although the DenseNet model might benefit from further fine-tuning to stabilize its validation performance. Figure 34 displays the training and validation performance of the EfficientNet model. The EfficientNet model demonstrates a rapid decline in training loss, indicating efficient learning. However, the validation loss is higher than the training loss and plateaus around 0.3, showing insufficient generalization in learning and overfitting. Training accuracy increases consistently, while validation accuracy also improves but with slight instability. This suggests that EfficientNet performs well but might be sensitive to specific variations in the validation dataset. Figure 35 illustrates the training and validation performance of the Xception model. The training loss decreases sharply during the initial epochs, reflecting effective optimization. However, the validation loss fluctuates, suggesting variability in the model's ability to generalize across different data samples. The training accuracy curve rises steadily, while validation accuracy shows significant oscillation. This behavior indicates that the Xception model captures features effectively but might need additional adjustments to handle data variability more consistently. Figure 36 presents the training and validation performance of the ResNet50 model. The training loss curve decreases steadily, highlighting consistent learning by the ResNet50 model. The validation loss also declines but remains around 0.75. The validation and training accuracy curves increase, while validation accuracy is lower than the training accuracy curve, showing that the model struggles to generalize well to unseen data.

Case study 6 (Apple disease detection)

Dataset

This dataset was downloaded from[59]. It was developed to train DL models to classify apple disease. This dataset has four classes: Apple scab, Black rot, Cedar apple rust, and healthy. The size of this dataset is 140.68 MB. It has two folders: train and test data. Each folder has four classes. It has 2016 images of Apple scab, 1987 images of Black rot, 1760 images of Cedar apple rust, and 2008 images of health. Figure 37 shows a sample of apple disease dataset.

In each folder, we applied IPP to the images. We resized the images into 224 and 224 with 3 channels. We put batch size with 256, label model is int, and labels are inferred. We show 7771 images in four classes in the training process and 1943 images in four classes in the testing process.

DL models

We applied the CNN model in this dataset to show the final prediction and better performance. Figure 38 shows the architecture of the CNN algorithm in apple disease classification. Then we apply the other four DL models. Table 6 shows the hyperparameters and results of DL models. Figures 39, 40, 41, 42 and 43 show the accuracy and loss for five DL models.

Figures 39, 40, 41, 42 and 43 illustrate the loss and accuracy plots for five DL models throughout the training and validation processes on the apple disease dataset. All five models performed well on this dataset. The training accuracy in Figs. 40, 41, 42 and 43 is slightly higher than the validation accuracy, showing overfitting or insufficient generalization. Also, all models show the validation loss is slightly higher than the training loss, indicating that the model performs better on the training data than the unseen validation data.

Discussions and analysis

This study applied deep learning (DL) models for the early detection of fruit plant diseases. These models can assist farmers and countries in quickly identifying and classifying fruit diseases, which is crucial for timely intervention. The study focused on six case studies, including apple, orange, mango, banana, grape, and guava diseases. We evaluated five DL models: CNN, DN, EN, XC, and RN. To improve accuracy and performance, we first applied image processing techniques and data augmentation. This included resizing images to various dimensions such as 224×224 and 256×256 pixels and performing other preprocessing steps like rescaling.

Orange disease detection

In the orange disease case study, the CNN model achieved the highest accuracy of 96.25%, followed closely by EN with 95.96%, and DN with 93.9%. RN and XC showed lower performance, with accuracies of 92.93%. In terms of training time, EN required the least amount of time (1361 s), followed by RN (1363 s). The longest training times were seen with XC (1898 s) and CNN (1501 s). EN also had the lowest loss (0.256), while XC exhibited the highest loss (12.5598).

Grape disease detection

For grape disease detection, the EN model achieved the highest accuracy of 99.14%, followed by RN with 98.15%, and DN with 96.55%. CNN and XC performed poorly, with accuracies of 92.36% and 83.48%, respectively. CNN had the shortest training time (1381 s), while EN took 3519 s. XC and RN required 4734 s and 3686 s, respectively. EN also had the lowest loss (0.0484), while XC had the highest loss (6.1152).

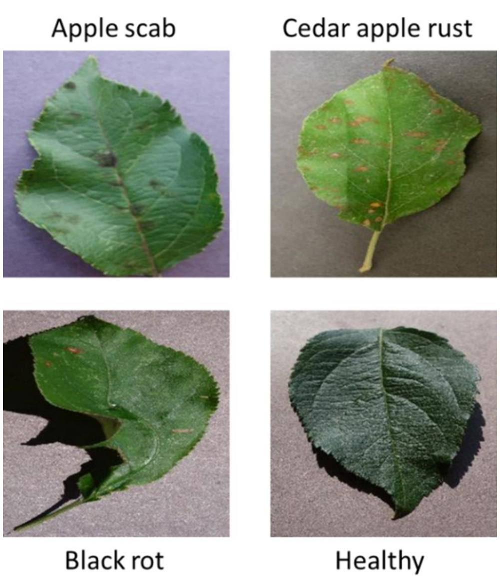

Fig. 37. Sample of apple disease.

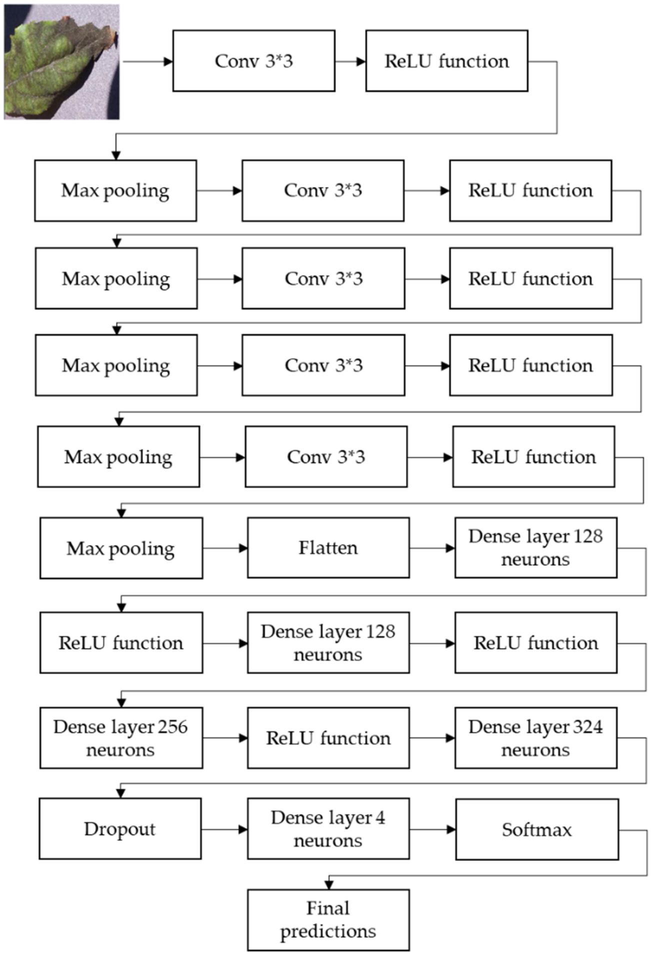

Fig. 38. The architecture of CNN algorithm.

	CNN	DN	EN	XC	RN
Opt	Adam				
LR	0.001				
Loss name	SCCE	SCCE	SCCE	SCCE	SCCE
Metrics	ACC	ACC	ACC	ACC	ACC
Batch size	256	256	256	256	256
Epoch	10	10	5	5	5
Time per step	58 s/step	29 s/step	25 s/step	34 s/step	25 s/step
Epoch time	1795 s	922 s	786 s	1032 s	788 s
Training time	17,953 s	9225 s	3390 s	5162 s	3940 s
Steps per epoch	31	31	31	31	31
ACC	92.52%	99.71%	99.84%	91.75%	100%
ACC validation	94.34%	97.48%	99.33%	85.64%	99.02%
Loss	0.2017	0.0084	0.0042	1.2794	0.00029
Loss validation	0.1692	0.1703	0.0261	3.5405	0.0728

Table 6. Hyperparameters and results of DL models in Apple disease.

Mango disease detection

In the mango disease dataset, the RN model achieved the highest accuracy (96.17%), followed by EN (95.83%) and DN (94.50%). CNN and XC showed the weakest performance, with accuracies of 76.83% and 70.33%, respectively. CNN had the shortest training time (1708 s), followed by EN at 2941 s. The longest training times

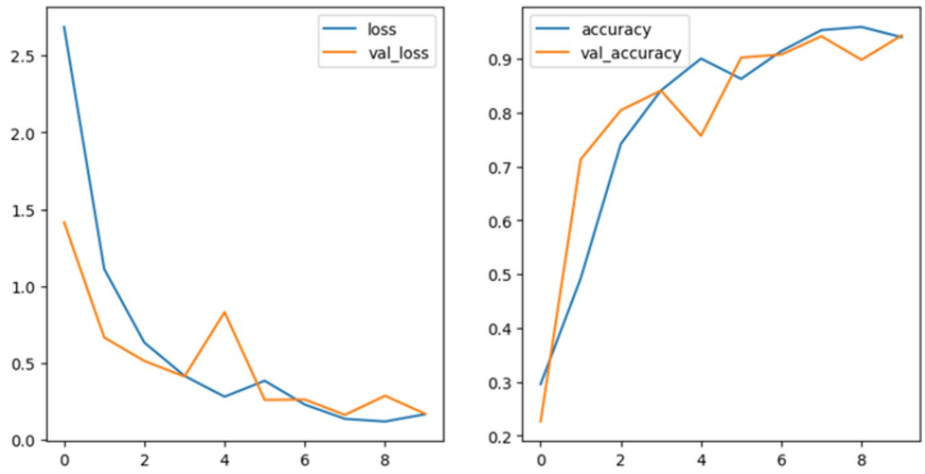

Fig. 39. Loss and accuracy for apple disease dataset by the CNN algorithm.

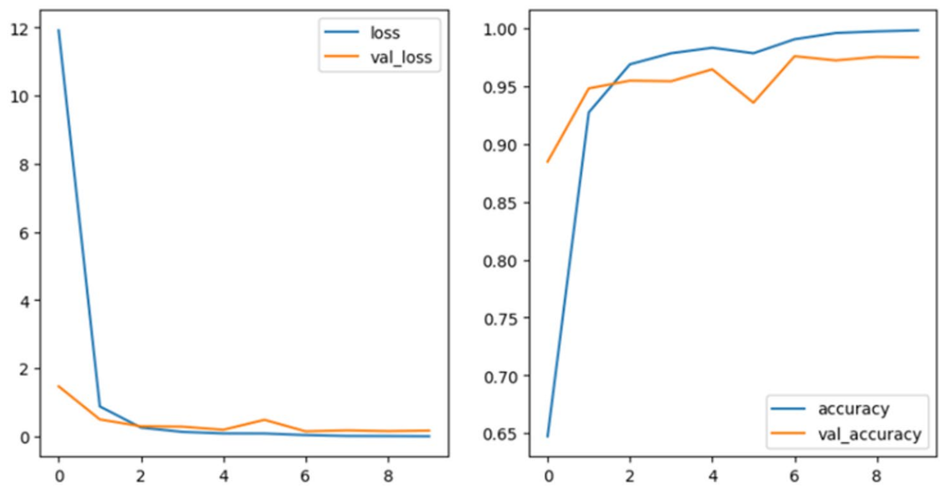

Fig. 40. Loss and accuracy for apple disease dataset by the DN algorithm.

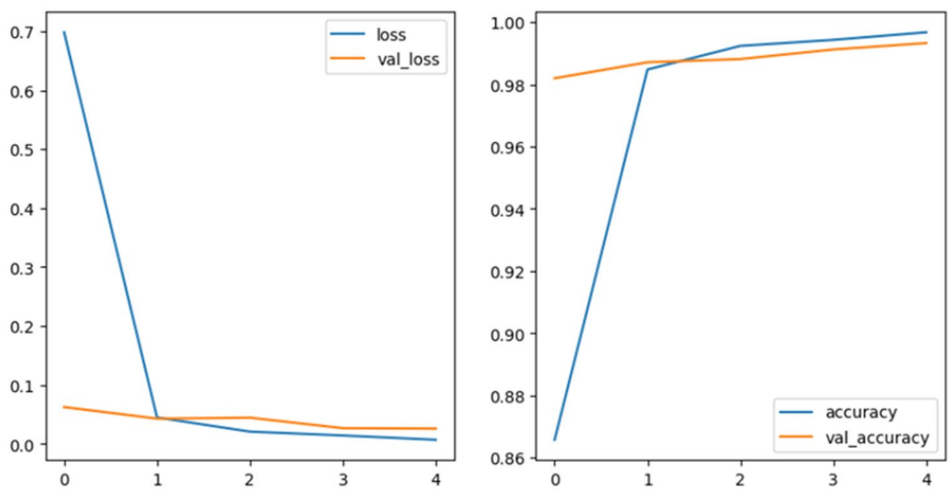

Fig. 41. Loss and accuracy for apple disease dataset by the EN algorithm.

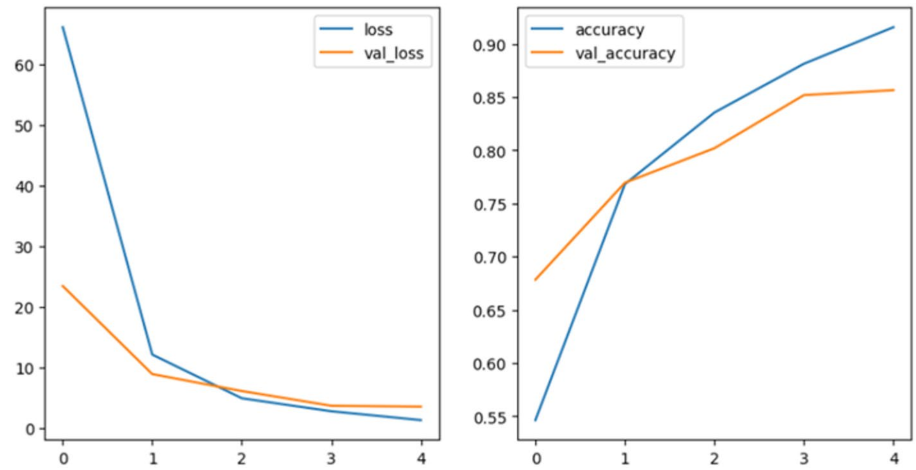

Fig. 42. Loss and accuracy for apple disease dataset by the XC algorithm.

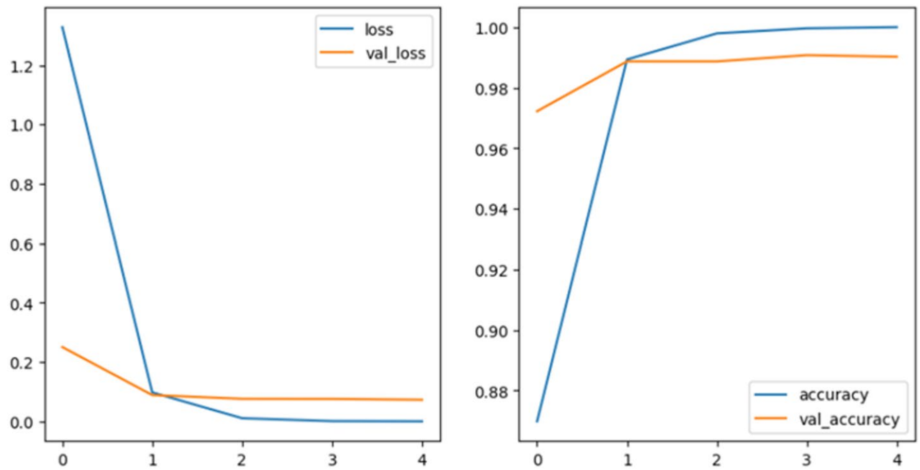

Fig. 43. Loss and accuracy for apple disease dataset by the RN algorithm.

were recorded for XC (3719 s) and RN (3170 s). EN had the lowest loss (0.5931), and CNN's loss was 0.6078, while XC had the highest loss (30.7733).

Banana disease detection

For the banana disease dataset, the DN model achieved the highest accuracy (94.06%), followed by XC with 92.50%. EN and RN performed the worst, with RN showing the fastest training time (617 s) and EN not far behind (627 s). CNN required 1280 s, and XC took 844 s. DN had the lowest loss (0.1806), followed by XC with 0.3820. EN exhibited the highest loss (1.8659).

Guava disease detection

For guava disease detection, RN achieved the highest accuracy (96.72%), followed closely by EN with 96.16%. CNN and XC were again the weakest performers, with accuracies of 53.28% and 75.3%, respectively. CNN had the shortest training time (1931 s), followed by RN (2165 s). The longest training time was observed for XC (2795 s), followed by DN (2326 s). EN achieved the lowest loss (0.2913), while RN had a loss of 0.6200. The highest loss was recorded for XC (22.7974).

Apple disease detection

In the apple disease dataset, the RN model achieved perfect accuracy (100%) in the training data, but EN obtained higher accuracy in the validation data. CNN and XC were the worst performing models, with accuracies of 94.34% and 85.64%, respectively. EN had the lowest loss (0.0261), followed by RN with 0.0728, while XC showed the highest loss (3.5405).

Ref	Model name	Fruits type	ACC
[60]	Alex net	Grape	99%
[60]	Alex net	Mango	89%
[61]	DNN	Mango	88.46%
[62]	SVM	Grape	88.89%
[63]	Ensemble DN, EN, and EN NoisyStuden	Apple	96.25%
[64]	CNN	Apple	98.54%
[65]	Faster R-CNN Inception V2	Banana	72.08%
[66]	SVM	Banana	85%
[67]	CNN	Guava	95.61
[68]	MobileNetV2	Guava	92.41%
[67]	SVM	Orange	82.3%
[69]	SVM	Orange	90%
Suggested models	CNN	Orange	96.25%
	EN	Grape	99.14%
	RN	Mango	96.17%
	DN	Banana	94.06%
	RN	Guava	96.72%
	RN	Apple	99.02%

Table 7. Comparative analysis between the proposed model and other works.

Comparative analysis
Table 7 provides a detailed comparison of the models used for fruit disease detection, highlighting the model names, fruit types, and their respective accuracies. The results show significant variation in model performance. Some models, like AlexNet for grape and EfficientNet for grape, achieved excellent accuracy (around 99%), while others, such as Faster R-CNN for banana, performed poorly with an accuracy of only 72.08%. The analysis also suggests that models like ResNet and EfficientNet hold considerable promise for even higher accuracy and efficiency, making them strong candidates for future implementations in fruit disease detection. This analysis is critical for the agricultural sector, as it provides insights into the most effective models for detecting diseases in specific fruits. Moreover, the study highlights how different machine learning techniques can improve the accuracy of fruit disease detection. The results from this comparative study are valuable for guiding farmers, researchers, and agricultural technology developers in selecting the most reliable and efficient models. By adopting these models, the agricultural industry can better protect crops, improve yields, and enhance food security.

Implications of our findings from deep learning models for agriculture
In this section, we discuss the implications of the findings from our deep learning (DL) models for agriculture. This study demonstrates the significant potential of DL models in improving the early detection of fruit plant diseases, which can be crucial for effective disease management and crop protection.

By applying five different DL models—CNN, DenseNet 121, EfficientNet B3, Xception, and ResNet 50—across six fruit disease datasets (orange, apple, banana, guava, grape, and mango), our findings show that these models can achieve high accuracy in disease classification. For example, EfficientNet B3 achieved an impressive accuracy of 99.14% in grape disease detection and 99.33% in apple disease detection, while ResNet 50 demonstrated strong performance with 96.17% accuracy in mango disease and 96.72% accuracy in guava disease. These results suggest that DL models can significantly improve the speed and accuracy of disease detection, which is essential for timely intervention. Early identification of diseases allows farmers to implement targeted treatments, reducing the risk of widespread crop damage and minimizing the use of pesticides, thereby supporting more sustainable agricultural practices.

Moreover, the efficiency of these models, in terms of both accuracy and training time, highlights their practical potential for real-time deployment in the field. Farmers can integrate these DL models into their daily operations, providing them with reliable tools to monitor plant health and manage diseases proactively. This capability can lead to higher yields, reduced costs, and ultimately more sustainable farming practices. the recommended deep learning models show great imply for enhancing agricultural practices, improving disease management, and contributing to food safety. By enabling faster, more accurate disease detection, these models can help farmers, researchers, and agricultural technologists to make better-informed decisions and protect crops more effectively.

Challenges in fruit plant disease management and their implications for agriculture
This section discusses the challenges that farmers face in dealing with fruit plant diseases and the implications these challenges have on agriculture.

One of the most significant challenges is the health and safety concerns associated with fruit diseases. Plant diseases not only threaten crop yield and quality but can also have detrimental effects on human health. Many fruit diseases can make fruits unsafe for consumption, leading to foodborne illnesses or contamination risks. Moreover, the use of chemical treatments for these diseases can further harm both the environment and public health. Therefore, ensuring the health of fruit plants is critical to maintaining food safety and preventing diseases from spreading to humans.

Another challenge is related to the effectiveness and performance of the models used for detecting and classifying fruit diseases. Detecting fruit diseases early is crucial to preventing their spread, but current diagnostic tools can be time-consuming, inaccurate, or inefficient. As fruit diseases can vary in appearance and affect different plant species differently, creating a reliable and precise detection model that works across various conditions remains a challenge. The complexity of these diseases and their ability to quickly spread through air, water, or human activity complicates early detection efforts.

Also, the economic impact of fruit diseases represents another significant challenge. If not detected and treated in time, plant diseases can spread quickly, leading to the destruction of entire crops. This not only reduces yield but also negatively impacts the quality of the fruit, making it unfit for export or consumption. The financial losses extend beyond the immediate loss of produce; farmers may face increased costs for pest control, disease management, and recovery efforts. Additionally, a decrease in fruit production can affect the local and national economy, especially in regions where agriculture is a major economic driver. fruit plant diseases pose a variety of challenges ranging from health risks to economic impacts. The key to overcoming these challenges lies in the development of efficient, accurate, and fast detection methods that can prevent the rapid spread of diseases, safeguard human health, and minimize financial losses for farmers and the agricultural industry as a whole.

Conclusions and future directions

This study proposed a deep learning (DL) model for fruit disease detection (FDD). We utilized five different DL models: CNN, DenseNet 121, EfficientNet B3, Xception, and ResNet 50. These models were applied to detect diseases in six different fruit crops: orange, apple, banana, guava, grape, and mango. The models were trained with various hyperparameters, and image preprocessing and data augmentation techniques were used to resize and enhance the images. We analyzed the performance of the models based on metrics such as accuracy, loss, training time, steps per epoch, and the number of epochs.

The results showed that the CNN model achieved the highest accuracy for detecting orange plant diseases (96.25%), while EfficientNet B3 performed best for grape disease detection (99.14%) and apple disease detection (99.33%). The ResNet 50 model achieved 96.17% accuracy for mango disease, DenseNet121 achieved 94.06% accuracy for banana disease, and ResNet 50 again performed well for guava disease with 96.72% accuracy. Overall, EfficientNet B3 was found to perform the best across several fruit disease datasets. These findings were compared with other deep learning models, demonstrating the effectiveness of the proposed models for FDD.

In future work, the scope of fruit disease detection can be expanded by incorporating a wider variety of fruit species into the datasets. By increasing the number of fruit types, the system can provide more comprehensive disease detection across different crops, which would be beneficial for farmers growing diverse types of fruit.

Moreover, exploring additional deep learning models and architectures could further improve the performance of FDD systems. Advanced models, such as Vision Transformers or hybrid approaches, could be tested to achieve higher accuracy and efficiency in detecting diseases, especially in more challenging cases.

Improving the image preprocessing techniques and data augmentation methods used in this study could also boost the quality of the input images. For example, incorporating more advanced image processing methods, such as adaptive histogram equalization or noise reduction, could enhance the clarity and detail of images, improving model performance. Expanding data augmentation strategies, such as rotating, flipping, or adding noise, would help the models generalize better and become more robust in real-world applications. These improvements would contribute to more effective and scalable fruit disease management systems, ultimately benefiting farmers, researchers, and agricultural industries. With enhanced detection capabilities, these systems could help prevent the spread of plant diseases, improve crop yields, and ensure better food quality.

Data availability

The datasets analysed during the current study are publicly available in the Kaggle repository. The sources of all datasets are cited in the References section (Refs. 59-64). No new datasets were generated during this study.

Received: 8 October 2025; Accepted: 29 January 2026
Published online: 10 February 2026

References

1. Singh, D. & Sharma, R. R. Postharvest diseases of fruits and vegetables and their management. In *Postharvest Disinfection of Fruits and Vegetables*, 1–52 (Elsevier, 2018).
2. Seeram, N. P. Berry fruits: compositional elements, biochemical activities, and the impact of their intake on human health, performance, and disease, *J. Agric. Food Chem.* **56**(3), 627–629 (2008).
3. Van't Veer, P., Jansen, M. C. J. F., Klerk, M. & Kok, F. J. Fruits and vegetables in the prevention of cancer and cardiovascular disease. *Public Health. Nutr.* **3** (1), 103–107 (2000).
4. Zhao, C. N. et al. Fruits for prevention and treatment of cardiovascular diseases. *Nutrients*. **9**(6), 598 (2017).
5. Mellendick, K. et al. Diets rich in fruits and vegetables are associated with lower cardiovascular disease risk in adolescents. *Nutrients*. **10** (2), 136 (2018).
6. Sandoval, J. R. M., Rosas, M. E. M., Sandoval, E. M., Velasco, M. M. M. & De Ávila, H. C. *Color Analysis and Image Processing Applied in Agriculture*, vol. 32 (InTech Rijeka, 2018).
7. Mushtaq, S. Smart agriculture system + and image processing. *Int. J. Adv. Res. Comput. Sci.* **9**, 1 (2018).

8. Raj, M. P., Swaminarayan, P. R. & Istar, A. Applications of image processing for grading agriculture products. *Int. J. Recent. Innov. Trends Comput. Commun*. **3** (3), 1194–1201 (2015).
9. Syeda, I. H., Alam, M. M., Illahi, U. & Su'ud, M. M. Advance control strategies using image processing, UAV and AI in agriculture: a review. *World J. Eng.* **18** (4), 579–589 (2021).
10. Santos, L., Santos, F. N., Oliveira, P. M. & Shinde, P. Deep learning applications in agriculture: A short review. In *Robot 2019: Fourth Iberian Robotics Conference: Advances in Robotics*, vol. 1, 139–151 (Springer, 2020).
11. Chen, H. et al. A deep learning CNN architecture applied in smart near-infrared analysis of water pollution for agricultural irrigation resources. *Agric. Water Manag.* **240**, 106303 (2020).
12. Pacal, I. et al. A systematic review of deep learning techniques for plant diseases. *Artif. Intell. Rev.* **57** (11), 304 (2024).
13. Bayram, B., Kunduracioglu, I., Ince, S. & Pacal, I. A systematic review of deep learning in MRI-based cerebral vascular occlusion-based brain diseases. *Neuroscience*. (2025).
14. Kunduracioglu, I. Cnn models approaches for robust classification of Apple diseases. *Comput. Decis. Mak. Int. J.* **1**, 235–251 (2024).
15. Kunduracioglu, I. & Pacal, I. Advancements in deep learning for accurate classification of grape leaves and diagnosis of grape diseases. *J. Plant. Dis. Prot.* **131** (3), 1061–1080 (2024).
16. Kunduracioglu, I. Utilizing ResNet architectures for identification of tomato diseases. *J. Intell. Decis. Mak. Inf. Sci.* **1**, 104–119 (2024).
17. İ, Kunduracıoğlu & Paçal, İ. Deep learning-based disease detection in sugarcane leaves: evaluating EfficientNet models. *J. Oper. Intell.* **2** (1), 235–321 (2024).
18. İ, Paçal & Kunduracıoğlu, İ. Data-efficient vision transformer models for robust classification of sugarcane. *J. Soft Comput. Decis. Anal.* **2** (1), 258–271 (2024).
19. Corceiro, A., Alibabaei, K., Assunção, E., Gaspar, P. & Pereira, N. Methods for detecting and classifying weeds, diseases and fruits using AI to improve the sustainability of agricultural crops: a review. *Processes*. **11**(4), 1263 (2023).
20. Bal, F. & Kayaalp, F. Review of machine learning and deep learning models in agriculture. *Int. Adv. Res. Eng. J.* **5** (2), 309–323 (2021).
21. Bharman, P. et al. Deep learning in agriculture: a review. *Asian J. Res. Comput. Sci.* **13** (2), 28–47 (2022).
22. Shireesha, G. & Reddy, B. Citrus fruit and leaf disease detection using DenseNet. In *2022 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)*, 1–5 (IEEE, 2022).
23. Kavitha, M., Suganthy, M., Srinivasan, R. & Kavitha, R. DenseNet-121 architecture for plant leaf disease classification. In *AIP Conference Proceedings*, vol. 2802, no. 1 (AIP Publishing, 2024).
24. Farman, H. et al. Efficientnet-based robust recognition of Peach plant diseases in field images. *Comput. Mater. Contin*. **71** (1), 2073–2089 (2022).
25. Atila, Ü., Uçar, M., Akyol, K. & Uçar, E. Plant leaf disease classification using EfficientNet deep learning model. *Ecol. Inform.* **61**, 101182 (2021).
26. Roopashree, S. & Anitha, J. DeepHerb: A vision based system for medicinal plants using Xception features. *IEEE Access.* **9**, 135927–135941 (2021).
27. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J. & Wojna, Z. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2818–2826 (2016).
28. Tanwar, V. & Lamba, S. Multiple grapes leaf disease identification using an optimal deep learning model: Xception. In *2023 2nd International Conference on Smart Technologies and Systems for Next Generation Computing (ICSTSN)*, 1–6 (IEEE, 2023).
29. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, 770–778 (2016).
30. Kumar, M., Arora, A., Deb, A. & Yadav, A. Deep learning for accurate plant disease classification using ResNet50: A comprehensive approach. In *2024 International Conference on Computational Intelligence and Computing Applications (ICCICA)*, vol. 1, 125–130 (IEEE, 2024).
31. Dhiman, P. et al. A novel deep learning model for detection of severity level of the disease in citrus fruits. *Electronics*. **11** (3), 495 (2022).
32. Nasir, I. M. et al. Deep learning-based classification of fruit diseases: an application for precision agriculture. *Comput. Mater. Contin*. **66** (2), 1949–1962 (2021).
33. Ashok, V. & Vinod, D. S. A novel fusion of deep learning and android application for real-time mango fruits disease detection. In *Intelligent System Design: Proceedings of Intelligent System Design: INDIA 2019*, 781–791 (Springer, 2021).
34. Gill, H. S. et al. Fruit recognition from images using deep learning applications. *Multimed. Tools Appl.* **81** (23), 33269–33290 (2022).
35. Assunçao, E., Diniz, C., Gaspar, P. D. & Proença, H. Decision-making support system for fruit diseases classification using Deep Learning. In *International Conference on Decision Aid Sciences and Application (DASA)*, 652–656 (IEEE, 2020).
36. Khattak, A. et al. Automatic detection of citrus fruit and leaves diseases using deep neural network model. *IEEE Access.* **9**, 112942–112954 (2021).
37. Gómez-Flores, W., Garza-Saldaña, J. J. & Varela-Fuentes, S. E. CitrusUAT: A dataset of orange citrus sinensis leaves for abnormality detection using image analysis techniques. *Data Brief.* **52**, 109908 (2024).
38. Banerjee, D., Sharma, N., Chauhan, R., Singh, M. & Kumar, B. V. Hybrid CNN & random forest model for effective bitter orange leaf disease diagnosis. In *4th International Conference on Innovative Practices in Technology and Management (ICIPTM)*, 1–6 (IEEE, 2024).
39. Dhiman, P. et al. Smart disease detection system for citrus fruits using deep learning with edge computing. *Sustainability* **15** (5), 4576 (2023).
40. Mojumdar, M. U. & Chakraborty, N. R. Orange leaves diseases detection using computerized techniques. In *12th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, 1–4 (IEEE, 2021).
41. Saha, R. & Neware, S. Orange fruit disease classification using deep learning approach. *Int. J.* **9**, 2 (2020).
42. Malathy, S., Karthiga, R. R., Swetha, K. & Preethi, G. Disease detection in fruits using image processing. In *6th International Conference on Inventive Computation Technologies (ICICT)*, 747–752 (IEEE, 2021).
43. Kukreja, V. & Dhiman, P. A deep neural network based disease detection scheme for citrus fruits. In *2020 International Conference on Smart Electronics and Communication (ICOSEC)*, 97–101 (IEEE, 2020).
44. Awate, A., Deshmankar, D., Amrutkar, G., Bagul, U. & Sonavane, S. Fruit disease detection using color, texture analysis and ANN. In *International Conference on Green Computing and Internet of Things (ICGCIoT)*, 970–975 (IEEE, 2015).
45. Gaikwad, S. A., Deore, K. S., Waykar, M. K., Dudhane, P. R. & Sorate, G. Fruit disease detection and classification. *Int. Res. J. Eng. Technol.* **4**, 1151–1154 (2017).
46. Agarwal, A., Sarkar, A. & Dubey, A. K. Computer vision-based fruit disease detection and classification. In *Smart Innovations in Communication and Computational Sciences: Proceedings of ICSICCS-2018*, 105–115 (Springer, 2019).
47. Shorten, C. & Khoshgoftaar, T. M. A survey on image data augmentation for deep learning. *J. Big Data*. **6** (1), 1–48 (2019).
48. Albelwi, S. A. Deep architecture based on DenseNet-121 model for weather image recognition. *Int. J. Adv. Comput. Sci. Appl.* **13**, 10 (2022).
49. Chhabra, M. & Kumar, R. A smart healthcare system based on classifier DenseNet 121 model to detect multiple diseases. In *Mobile Radio Communications and 5G Networks: Proceedings of Second MRCN 2021*, 297–312 (Springer, 2022).

50. Tan, M. & Le, Q. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning*, 6105–6114 (PMLR, 2019).
51. Batool, A. & Byun, Y. C. Lightweight EfficientNetB3 model based on depthwise separable convolutions for enhancing classification of leukemia white blood cell images. *IEEE Access.* **11**, 37203–37215 (2023).
52. Chollet, F. Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 1251–1258 (2017).
53. Theckedath, D. & Sedamkar, R. R. Detecting affect states using VGG16, ResNet50 and SE-ResNet50 networks. *SN Comput. Sci.* **1** (2), 79 (2020).
54. Jonathan & Cristovão Orange diseases dataset. Accessed 01 December 2024 (2024). https://www.kaggle.com/datasets/jonathansilva2020/orange-diseases-dataset
55. Rajarshi, M. Grapevine Disease Dataset. Accessed 01 December 2024 (2024). https://www.kaggle.com/datasets/rm1000/grape-disease-dataset-original
56. Arya, S. Mango Leaf Disease Dataset. Accessed 01 December 2024 (2022). https://www.kaggle.com/datasets/aryashah2k/mango-leaf-disease-dataset
57. Shifat, E. A. Banana Leaf Spot Diseases (BananaLSD) Dataset (2023). https://www.kaggle.com/datasets/shifatearman/bananalsd (accessed 01 Dec 2024).
58. Jaswanth, K. E. Guava Disease Dataset. (2023). https://www.kaggle.com/datasets/jaswanthkrishnaeaga/guava-disease-dataset (accessed 01 Dec 2024).
59. Md Rashedul Alam Anik. Apple Disease Dataset. (2021). https://www.kaggle.com/datasets/ludehsar/apple-disease-dataset (accessed 01 Dec 2024).
60. Rao, U. S. et al. Deep learning precision farming: grapes and mango leaf disease detection by transfer learning. *Glob. Transit. Proc.* **2**(2), 535–544 (2021).
61. Trang, K., TonThat, L., Thao, N. G. M. & Thi, N. T. T. Mango diseases identification by a deep residual network with contrast enhancement and transfer learning. In *IEEE Conference on Sustainable Utilization and Development in Engineering and Technologies (CSUDET)*, 138–142 (IEEE, 2019).
62. Padol, P. B. & Yadav, A. A. SVM classifier based grape leaf disease detection. In *Conference on Advances in Signal Processing (CASP)*, 175–179 (IEEE, 2016).
63. Bansal, P., Kumar, R. & Kumar, S. Deep residual learning for image recognition leaves using deep convolutional neural network. *Agriculture* **11** (7), 617 (2021).
64. Alqethami, S., Almtanni, B., Alzhrani, W. & Alghamdi, M. Disease detection in Apple leaves using image processing techniques. *Eng. Technol. Appl. Sci. Res.* **12** (2), 8335–8341 (2022).
65. Selvaraj, M. G. et al. AI-powered banana diseases and pest detection. *Plant. Methods*. **15**, 1–11 (2019).
66. Chaudhari, V. & Patil, M. Banana leaf disease detection using K-means clustering and Feature extraction techniques. In *International Conference on Advances in Computing, Communication & Materials (ICACCM)*, 126–130 (IEEE, 2020).
67. Al Haque, A. S. M. F., Hafiz, R., Hakim, M. A. & Islam, G. M. R. A computer vision system for guava disease detection and recommend curative solution using deep learning approach. In *22nd International Conference on Computer and Information Technology (ICCIT)*, 1–6 (IEEE, 2019).
68. Rashid, J., Khan, I., Ali, G., Alturise, F. & Alkhalifah, T. Real-Time multiple guava leaf disease detection from a single leaf using hybrid deep learning technique. *Comput. Mater. Continua*. **74**, 1 (2023).
69. Behera, S. K., Jena, L., Rath, A. K. & Sethy, P. K. Disease classification and grading of orange using machine learning and fuzzy logic. In *International Conference on Communication and Signal Processing (ICCSP)*, 678–682 (IEEE, 2018).
70. Kamilaris, A. & Prenafeta-Boldú, F. X. Deep learning in agriculture: A survey. *Comput. Electron. Agric.* **147**, 70–90 (2018).
71. Latif, G., Abdelhamid, S. E., Mallouhy, R. E., Alghazo, J. & Kazimi, Z. A. Deep learning utilization in agriculture: detection of rice plant diseases using an improved CNN model. *Plants*. **11** (17), 2230 (2022).

Acknowledgements
This publication was made possible by the support of the AUK Open Access Publishing Fund.

Author contributions
All authors contribute the same in this study and reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to M.S.

Reprints and permissions information is available at www.nature.com/reprints.